UNSTEADY-STATE PRESSURE DISTRIBUTIONS CREATED BY A WELL WITH A SINGLE HORIZONTAL FRACTURE, PARTIAL PENETRATION, OR RESTRICTED ENTRY

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Dedicated *to* Emmanuel

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ABSTRACT

Although there have been many studies on unsteady behavior of wells with vertical fractures, and although there was at one time a controversy concerning the occurrence of horizontal or vertical fractures as a result of hydraulic fracturing -- there has been ~~no~~ study of the unsteady behavior of a well containing a horizontal fracture published to date. This is particularly surprising, because such a study might have indicated significant differences between performance for wells with horizontal and vertical fractures. The purpose of this study was to fill this existing gap in knowledge of fractured well behavior.

Initially, a finite-element model of a horizontal fracture was used to obtain behavior. Unfortunately, this model was not capable of producing accurate results at early producing times. Consequently, an analytical solution was developed by means of the concept of the instantaneous point source. The analytical solution modelled behavior of constant rate production from a well containing a single, horizontal fracture of finite thickness at any position within a producing interval in an infinitely large reservoir with impermeable upper and lower boundaries. This general solution also contained solutions for the case of: **(1)** a single, plane (zero thickness) horizontal fracture, (2) partial penetration of the producing formation, and **(3)** limited flow entry throughout a producing interval. Although these are interesting solutions, the main purpose of this study was investigation of the horizontal fracture case. The analytical solution for this case

was evaluated by the computer to produce tables of dimensionless pressures vs. dimensionless times sufficient for well-test analysis purposes.

A careful analysis of the general solution for a horizontal fracture indicated the existence of four different flow periods. During the first period, it appears that all production originates within the fracture causing a typical storage-controlled period. This period is followed by a period of vertical, linear flow. There then follows a transitional period, after which flow appears essentially radial. During this period, the pressure is the same as that created by a line source well with a skin effect. The skin effect is independent of time, but does depend upon the position of the pressure point. It was also found that there is a radius of influence beyond which flow is essentially radial for all times. Approximating solutions and appropriate time limits for approximate solutions were derived,

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INTRODUCTION

Hydraulic fracturing has been used for improving well productivity for the last twenty years, and is generally recognized as a major development in well completion technology. There was considerable discussion in the early 1950's about the orientation and the number of fractures created by this type of well stimulation. It is generally agreed[1] from theoretical considerations, that a vertical fracture will result if the least principal stress in the formarion is horizontal, whereas a horizontal fracture will be created if the least principal stress is vertical. However, data collected and reported by Zemanek, et al.[2], shows that hydraulic fracturing usually results in one vertical fracture, the plane of which includes the axis of the wellbore, This conclusion appears widely held today. Thus, most studies of the flow behavior for fractured wells consider vertical fractures only.

Many authors have studied the probiem of pressure behavior for wells with vertical fractures, either analyticaily[3,4,5], experimentally[6,7,8], or numerically[9,10]. Generally, it has been assumed that the pressure in the fracture is the same everywhere, and that the fracture offers no resistance to flow. Both, infinite[5] and finite reservoir sizes were considered, with either a circular[3,4,10], or square[9] outer boundary; and solutions have been extended to ideal[11] and real[10] gas flow.

The early time behavior of the pressure is the same as if the flow were linear: the linear flow period may then be followed, after a period of transition, by a radial-type flow period[5]. Apparent skin factors have been caiculared from which it appears that a vertically-fractured well can be approximated by an unfractured well with an equivalent wellbore radius equal to half the well axis to the fracture tip length[3]. Methods for analyzing pressure build-up data have been proposed by Russel and Truitt[9], who suggested a Muskat-type plot[12] for determining average pressure, and by Ramey for early time data[10,13]

The pseudo-steady state behavior of horizontally-fractured wells has been studied numerically by Hartsock and Warren[114]. Their model assumed the reservoir to be homogeneous, of constant: thickness, of anisotropic permeabilities and completely penetrated by a well of small radius. A single, horizontal, symmetrical fracture of negligible thickness and finite conductivity was located at the center of the formation. Radial flow was assumed beyond a critical radius four times as large as the fracture radius, and there was no flow across the drainage radius. The only flow into the well itself was via the fracture. With this model, skin factors that were independent of the drainage radius could be calculated for various combinations of parameters. These authors concluded that a poorly-designed fracture treatment could yield a reduction in productivity, and that the apparent skin factor curves could be used in fracture treatment design.

Further knowledge of the transient pressure behavior for horizontal fractures of infinite conductivity located at the center, top, or bottom of the formation was extracted by Warren" in his discussion of Coats,[16] mathematical model for water movement about bottom-water-drive reservoirs. Warren gave an expression for the skin factor that depended upon the fracture radius, the well radius, the reservoir thickness, the ratio of the horizontal permeability to the vertical permeability, and a geometrical constant obtainable from Coats' results. No study of the unsteady-state behavior of wells with horizontal fractures has yet appeared.

It is the purpose of the present study to develop a mathematical model for investigating the unsteady state behavior of horizontally fractured wells. The solution has been found to be applicable to plane horizontal fractures, and to wells with partial penetration or limited flow entry. For each case, an extensive study of the dimensionless pressure drop function has been made, and different flow periods have been recognized.

I. UNSTEADY-STATE PRESSURE DISTRIBUTION CREATED BY A SINGLE HORIZONTAL FRACTURE OF THICKNESS h_f

A. MATHEMATICAL MODEL

The transient flow of a slightly compressible fluid in a porous medium is described by the diffusivity equation derived from the continuity equation and Darcy's law[17]. Limiting flow to the r and z directions, this equation is written in cylindrical coordinates as[18]:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\,\frac{\partial p(r,z,t)}{\partial r}\right)+\frac{k_z}{k_r}\,\frac{\partial^2 p(r,z,t)}{\partial z^2}=\frac{\phi\mu c}{k_r}\,\frac{\partial p(r,z,t)}{\partial t} \tag{1}$$

when the effect of gravity is neglected, and the permeabilities and fluid viscosity are assumed constant.

The ratio $\frac{k_z}{k_r}$ accounts for the anisotropy of the porous medium, and can be incorporated into the vertical coordinate by setting:

$$z = \hat{z}\sqrt{\frac{k_r}{k_z}} \tag{2}$$

If we denote

$$\eta = \frac{k_r}{\phi\mu c}$$

Eq. 1 can be written as

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\,\frac{\partial p(r,z,t)}{\partial r}\right)+\frac{\partial^2 p(r,z,t)}{\partial z^2}=\frac{1}{\eta}\,\frac{\partial p(r,z,t)}{\partial t} \tag{4}$$

The solution of a particular flow problem is completely determined by the initial and boundary conditions.

Many analytical and numerical techniques have been developed for solving such problems. Most solution techniques were first used for solving heat conduction problems, and have since been applied to petroleum engineering by many different authors.

One useful method employs Lord Kelvin's'' instantaneous point source solution[20].

$$\Delta p(d,t) = \frac{q}{8\phi c(\pi\eta t)^{3/2}} \exp\left(-\frac{d^2}{4\eta t}\right) \quad (5)$$

It can e verified that Eq. 5 s a solution of Eq. 4. Eq. 5 represents the pressure drop created at a point M by an instantaneous point source P of strength **q**; d is the distance PM. The pressure distribution created by simple instantaneous volumetric sources can be obtained by integrating the right hand side of Eq. 5 over the volume of the source. In this case, q is the instantaneous withdrawal rate per unit volume. **A** continuous volumetric source may then be obtained by integrating the instantaneous solution with respect to time.

This method of generating the pressure distribution created by volumetric sources is an example of superposition in time and space of independent solutions to a linear differential equation. The method was first applied in petroleum engineering by Nisle[21]. We will use this method for deriving the pressure distribution created at any point in the reservoir by a single horizontal fracture.

In order to simplify the mathematical definition of this problem, we will make the following physical assumptions:

1. We consider a horizontal, anisotropic, homogeneous reservoir of thickness h, porosity ϕ, with radial and vertical. permeabilities k_r and k_z, respectively. The reservoir is of infinite extent, and completely penetrated by a well of infinitely small radius, r_w.

2. A single, horizontal, symmetrical fracture of radius r_f, thickness h_f, and infinite conductivity is centered at the well, and the horizontal plane of symmetry of the fracture is at an altitude z_f.

3. A single-phase, slightly compressible liquid flows from the reservoir into the fracture at a constant rate q_f, which is uniform over the entire volume of the fracture.

4. There is no flow across the upper and Power boundaries of the reservoir, and the pressure remains unchanged and equal to the initial pressure as radial distance approaches infinity. A cross-section of the reservoir-fracture model is shown in Fig. 1.

We first consider an infinitesimal element of fracture, of volume $r dr dz d\theta$, as shown in Fig. 2, and calculate the pressure distribution at any point of an infinite medium. The result is given by:

$$\Delta p^{(3)} = \frac{q\ r\ dr\ d\theta\ dz}{8\phi c\ (\pi\eta t)3/2} \quad \exp\left(-\frac{d^2}{4\eta t}\right) \tag{6}$$

Because

$$d^2 = PP'^2 + P'M^2$$

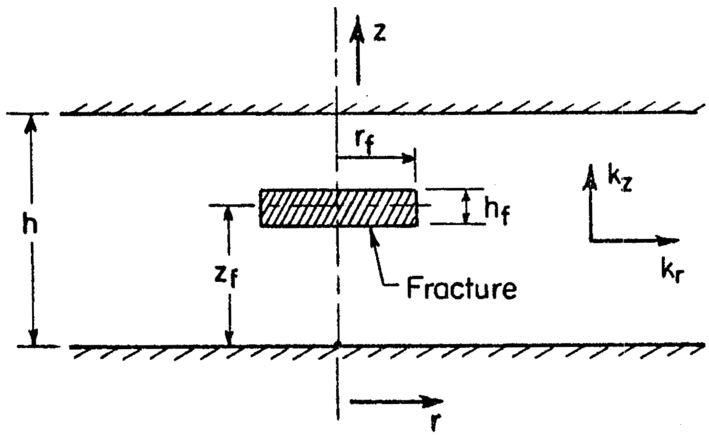

Fig. 1. **Cross** section of horizontal fracture model

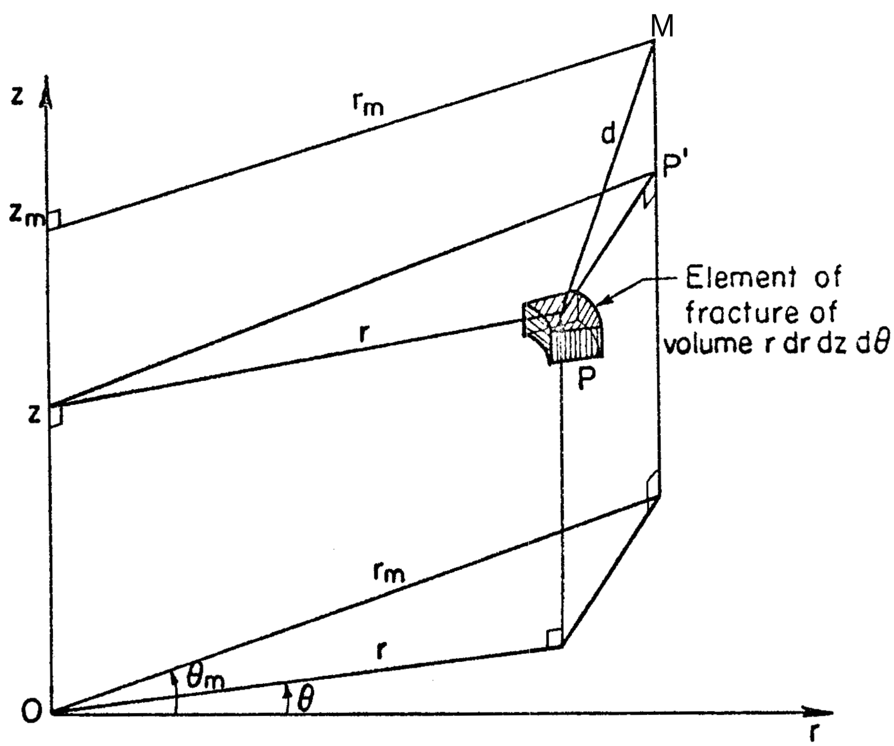

Fig. **2.** Schematic of an element of fracture volume in an infinite medium and

$$PP'^2 = r^2 + r_m^2 - 2rr_m \cos(\theta_m - \theta)$$

$$P'M^2 = (z_m - z)^2$$

then

$$d^2 = (z_m - z)^2 + r_m^2 + r^2 - 2rr_m \cos(\theta_m - \theta)$$

r_m, z_m and θ_m are constant, and correspond to the point M at which we calculate the pressure. r, z, and θ correspond to the element of source P, and vary to generate the fracture.

We first generate a ring source of radius r, at the altitude z, by integrating the right hand side of **Eq.** *6* with respect to θ from 0 to 2π to obtain:

$$\Delta p^{(2)} = \frac{q\ r\ dr\ dz}{8\phi c(\pi\eta t)^{3/2}} \exp\left[- \frac{(z - z_m)^2 + r_m^2 + r^2}{4\eta t}\right]$$

$$\int_0^{2\pi} \exp\left[\frac{r_m\ r \cos(\theta_m - \theta)}{2\eta t}\right] d\theta$$

From the definition of the modified Bessel function of the first kind of zero order, this result can be written as:

$$\Delta p^{(2)} = \frac{q\ r\ dr\ dz}{4\sqrt{\pi\phi c}(\eta t)^{3/2}} \left[- \frac{(z - z_m)^2 + r_m^2 + r^2}{4\eta t} \right] I_o \left(\frac{r_m r}{2\eta t} \right) \qquad (7)$$

The pressure distribution created by a finite height instantaneous Cylindrical surface of radius r and height h_f, whose horizontal plane of symmetry is at the altitude z_f, is obtained by integrating Eq. 7 with respect to z from $(z_f - \frac{h_f}{2})$ to $(z_f + \frac{h_f}{2})$. This leads to:

$$\Delta p^{(1)} = \frac{q\ r\ dr}{4\sqrt{\pi\phi c}(\eta t)^{3/2}} \exp \left(- \frac{r_m^2 + r^2}{4\eta t} \right) I_o \left(\frac{r_m r}{2\eta t} \right)$$

$$\int_{z_f - \frac{h_f}{2}}^{z_f + \frac{h_f}{2}} e^{- \frac{(z - z_m)^2}{4\eta t}} dz \qquad (8)$$

By using $u = - \frac{z - z_m}{2\sqrt{\eta t}}$, the integral can be written as:

$$2\sqrt{\eta t} \int_{\frac{z_m - z_f - \frac{h_f}{2}}{2\sqrt{\eta t}}}^{\frac{z_m - z_f + \frac{h_f}{2}}{2\sqrt{\eta t}}} e^{-u^2} du$$

From definition of the error function[22], this is:

$$\sqrt{\pi\eta t} \left[\operatorname{erf} \left(\frac{z_m - z_f + \frac{h_f}{2}}{2\sqrt{\eta t}} \right) - \operatorname{erf} \left(\frac{z_m - z_f - \frac{h_f}{2}}{2\sqrt{\eta t}} \right) \right]$$

and Eq. 8 becomes:

$$\Delta p^{(1)} = \frac{q\ r\ dr}{4\phi c\eta t} \left[\mathrm{erf}\left(\frac{z_m - z_f + \frac{h_f}{2}}{2\sqrt{\eta t}} \right) - \mathrm{erf}\left(\frac{z_m - a_f - \frac{h_f}{2}}{2\sqrt{\eta t}} \right) \right]$$

$$I_o \left(\frac{r_m r}{2\eta t} \right) \exp\left(- \frac{r_m^2 + r^2}{4\eta t} \right) \tag{9}$$

The instantaneous horizontal fracture is then obtained by integrating the right hand side of Eq. 9 with respect to r, from 0 to r_f; the final result is:

$$\Delta p = \frac{q\ e^{-\frac{r_m^2}{4\eta t}}}{4\phi c\eta t} \left[\mathrm{erf}\left(\frac{z_m - z_f + \frac{h_f}{2}}{2\sqrt{\eta t}} \right) - \mathrm{erf}\left(\frac{z_m - z_f - \frac{h_f}{2}}{2\sqrt{\eta t}} \right) \right]$$

$$\int_0^{r_f} I_o \left(\frac{r_m r}{2\eta t} \right) e^{-\frac{r^2}{4\eta t}}\ r\ dr \tag{10}$$

We will now use the method of images[23] to generate the upper and lower no-flow boundaries. We first generate the lower boundary by adding another identical fracture with the same production rate, symmetrically with respect to the lower boundary. The pressure drop created by the fracture and its image at the point M of the infinite medium is identical to the pressure drop created by the fracture at the point M in the semi infinite medium bounded by the lower boundary. The fracture and its image are shown on Fig. 3. The pressure drop at

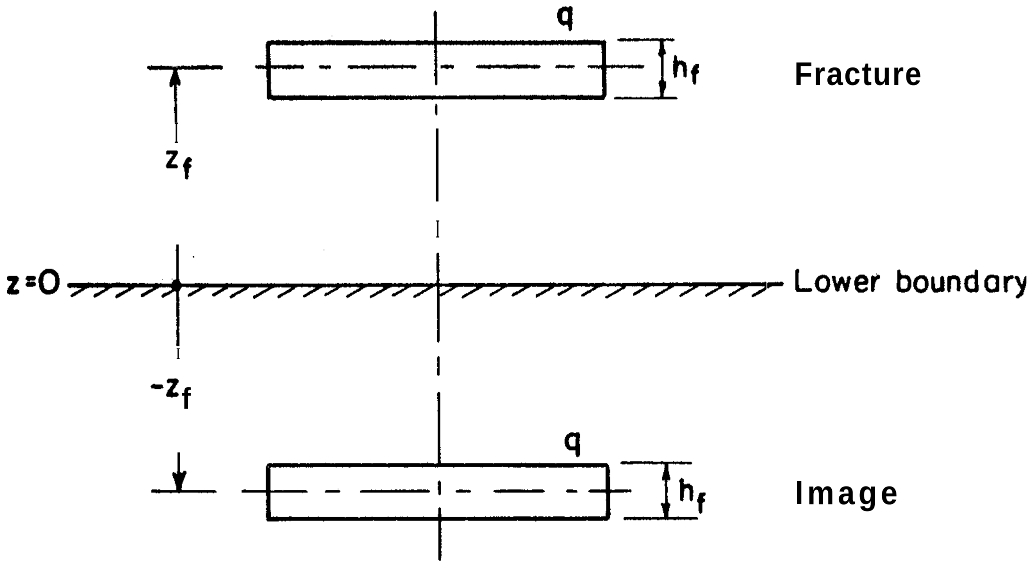

Fig. 3. Generation of lower boundary by method of images

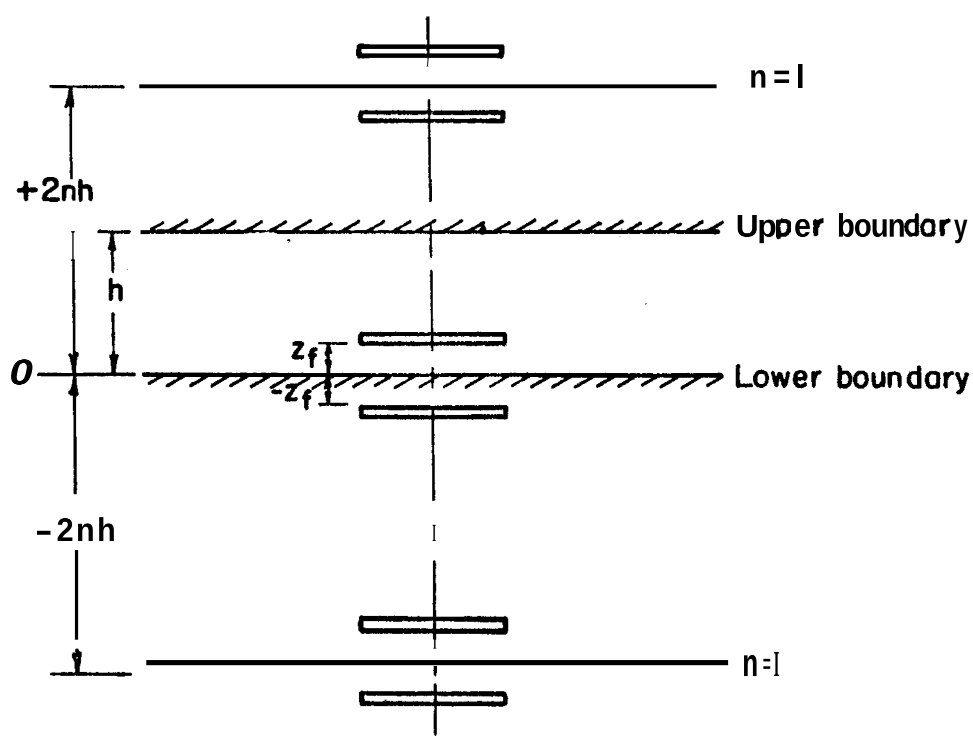

Fig. 4. Complete system of images to obtain upper and lower no-flow boundaries

M is the sum of the pressure drops from the two fractures, therefore:

$$\Delta p = \frac{e^{-\frac{r_m^2}{4\eta t}}}{4\phi c\eta t}\left[\operatorname{erf}\left(\frac{z_m - z_f + \frac{h_f}{2}}{2\sqrt{\eta t}}\right) - \operatorname{erf}\left(\frac{z_m - z_f - \frac{h_f}{2}}{2\sqrt{\eta t}}\right)\right.$$

$$\left. + \operatorname{erf}\left(\frac{z_m + z_f + \frac{h_f}{2}}{2\sqrt{\eta t}}\right) - \operatorname{erf}\left(\frac{z_m + z_f - \frac{h_f}{2}}{2\sqrt{\eta t}}\right)\right]$$

$$\int_0^{r_f} I_o\left(\frac{r\, r_m}{2\eta t}\right) e^{-\frac{r^2}{4\eta t}}\, r\, dr \qquad (11)$$

The upper boundary will be generated in a similar manner; but, this time, we must consider the images of the <u>two</u> fractures with respect to the upper boundary. These require two images which in turn require two images with respect to the lower boundary, and so on. We thus obtain an infinite number of couple of fractures, symmetrical with respect to planes located at +2nh and - 2nh from the lower boundary, where n varies from 1 to infinity. The total pressure drop created at M by these fractures is equal to the pressure drop created at M by the single horizontal fracture between the upper and lower reservoir boundaries. The system of images is shown in Fig. 41.

The pressure drop at M caused by instantaneous fractures is:

$$\Delta p = \frac{q\, e^{-\frac{r_m^2}{4\eta t}}}{4\phi c \eta t} \left\{ \operatorname{erf}\left(\frac{z_m - z_f + \frac{h_f}{2}}{2\sqrt{\eta t}}\right) - \operatorname{erf}\left(\frac{z_m - z_f - \frac{h_f}{2}}{2\sqrt{\eta t}}\right) \right.$$

$$+ \operatorname{erf}\left(\frac{z_m + z_f + \frac{h_f}{2}}{2\sqrt{\eta t}}\right) - \operatorname{erf}\left(\frac{z_m + z_f - \frac{h_f}{2}}{2\sqrt{\eta t}}\right)$$

$$+ \sum_{n=1}^{\infty} \left[\operatorname{erf}\left(\frac{z_m + 2nh - z_f + \frac{h_f}{2}}{2\sqrt{\eta t}}\right) - \operatorname{erf}\left(\frac{z_m + 2nh - z_f - \frac{h_f}{2}}{2\sqrt{\eta t}}\right) \right.$$

$$+ \operatorname{erf}\left(\frac{z_m + 2nh + z_f + \frac{h_f}{2}}{2\sqrt{\eta t}}\right) - \operatorname{erf}\left(\frac{z_m + 2nh + z_f - \frac{h_f}{2}}{2\sqrt{\eta t}}\right)$$

$$+ \operatorname{erf}\left(\frac{z_m - 2nh - z_f + \frac{h_f}{2}}{2\sqrt{\eta t}}\right) - \operatorname{erf}\left(\frac{z_m - 2nh - z_f - \frac{h_f}{2}}{2\sqrt{\eta t}}\right)$$

$$\left.\left. + \operatorname{erf}\left(\frac{z_m - 2nh + z_f + \frac{h_f}{2}}{2\sqrt{\eta t}}\right) - \operatorname{erf}\left(\frac{z_m - 2nh + z_f - \frac{h_f}{2}}{2\sqrt{\eta t}}\right) \right] \right\}$$

$$\int_0^{r_f} I_o\left(\frac{r_m r}{2\eta t}\right) e^{-\frac{r^2}{4\eta t}}\, r\, dr$$

This expression can be rearranged to give:

$$\Delta p = \frac{q\, e^{-\frac{r_m^2}{4\eta t}}}{4\phi c \eta t} \sum_{n=-\infty}^{n=+\infty} \left\{ \operatorname{erf} \frac{(z_m + 2nh) - (z_f - \frac{h_f}{2})}{2\sqrt{\eta t}} - \operatorname{erf} \frac{(z_m + 2nh) + (z_f - \frac{h_f}{2})}{2\sqrt{\eta t}} - \operatorname{erf} \frac{(z_m + 2nh) - (z_f + \frac{h_f}{2})}{2\sqrt{\eta t}} + \operatorname{erf} \frac{(z_m + 2nh) + (z_f + \frac{h_f}{2})}{2\sqrt{\eta t}} \right\} \int_0^{r_f} I_o\left(\frac{r_m r}{2\eta t}\right) e^{-\frac{r^2}{4\eta t}}\, r\, dr \qquad (12)$$

From the definition of the error function[21], we have:

$$\sum_{n=-\infty}^{+\infty} \left\{ \operatorname{erf} \frac{(z_m + 2nh) + (z_f + \frac{h_f}{2})}{2\sqrt{\eta t}} - \operatorname{erf} \frac{(z_m + 2nh) - (z_f + \frac{h_f}{2})}{2\sqrt{\eta t}} \right\}$$

$$= \frac{2}{\sqrt{\pi}} \sum_{n=-\infty}^{+\infty} \int_{\frac{(z_m + 2nh) - (z_f + \frac{h_f}{2})}{2\sqrt{\eta t}}}^{\frac{(z_m + 2nh) + (z_f + \frac{h_f}{2})}{2\sqrt{\eta t}}} e^{-u^2}\, du \qquad (13)$$

By changing the variable in the integrand, the right hand side of Eq. 13 becomes :

$$\frac{1}{\quad} \sum_{n=-\infty}^{+\infty} \int_{-(z_f + \frac{h_f}{2})}^{z_f + \frac{h_f}{2}} \exp\left[- \frac{(z_m - z + 2nh)^2}{4\eta t}\right] dz$$

which is equal *to:*

$$\frac{1}{\sqrt{\pi\eta t}} \int_{-(z_f + \frac{h_f}{2})}^{z_f + \frac{h_f}{2}} \sum_{n=-\infty}^{+\infty} \exp\left[- \frac{(z - z_m - 2nh)^2}{4\eta t}\right] dz$$

From Poisson's summation formula24..

$$\sum_{n=-\infty}^{+\infty} \exp\left[- \frac{(z - z_m - 2nh)^2}{4\eta t}\right]$$

$$= \frac{\sqrt{\pi\eta t}}{h}\left[1 + 2\sum_{n=1}^{\infty} \cos\frac{n\pi(z - z_m)}{h} \exp\left(- \frac{n^2\pi^2\eta t}{h^2}\right)\right]$$

Therefore,

$$\sum_{n=-\infty}^{n=+\infty}\left\{\operatorname{erf}\frac{(z_m + 2nh) + (z_f + \frac{h_f}{2})}{2\sqrt{\eta t}} - \operatorname{erf}\frac{(z_m + 2nh) - (z_f + \frac{h_f}{2})}{2\sqrt{\eta t}}\right\} =$$

$$\frac{1}{h}\left\{2\left(z_f + \frac{h_f}{2}\right) + 2 \sum_{n=1}^{\infty} \exp\left(- \frac{n^2\pi^2\eta t}{h^2}\right) \int_{-(z_f + \frac{h_f}{2})}^{z_f + \frac{h_f}{2}} \cos \frac{n\pi(z - z_m)}{h} \, dz\right\} \tag{14}$$

and

$$\int_{-(z_f + \frac{h_f}{2})}^{z_f + \frac{h_f}{2}} \cos \frac{n\pi(z - z_m)}{h} \, dz = \frac{h}{n\pi} \left[\sin n\pi \frac{z_f - z_m + \frac{h_f}{2}}{h} + \sin n\pi \frac{z_f + z_m + \frac{h_f}{2}}{h}\right] = \frac{2h}{n\pi} \sin \frac{n\pi}{h} \left(z_f + \frac{h_f}{2}\right) \cos \frac{n\pi z_m}{h}$$

Hence, Eq. 14 becomes:

$$\sum_{n=-\infty}^{+\infty} \left\{\operatorname{erf} \frac{(z_m + 2nh) + (z_f + \frac{h_f}{2})}{2\sqrt{\eta t}} - \operatorname{erf} \frac{(z_m + 2nh) - (z_f + \frac{h_f}{2})}{2\sqrt{\eta t}}\right\}$$

$$= \frac{1}{h}\left\{2\left(z_f + \frac{h_f}{2}\right) + \frac{4h}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \exp\left(- \frac{n^2\pi^2\eta t}{h^2}\right) \sin \frac{n\pi}{h} \left(z_f + \frac{h_f}{2}\right) \cos \frac{n\pi z_m}{h}\right\} \tag{15}$$

We can prove in a similar manner that:

$$\sum_{n=-\infty}^{+\infty}\left\{\operatorname{erf}\frac{(z_m + 2nh) - (z_f - \frac{h_f}{2})}{2\sqrt{\eta t}} - \operatorname{erf}\frac{(z_m + 2nh) + (z_f - \frac{h_f}{2})}{2\sqrt{\eta t}}\right\}$$

$$= \frac{-1}{h}\left\{2(z_f + \frac{h_f}{2}) + \frac{4h}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\exp\left(-\frac{n^2\pi^2\eta t}{h^2}\right)\right.$$

$$\left.\sin\frac{n\pi}{h}(z_f - \frac{h_f}{2})\cos\frac{n\pi z_m}{h}\right\} \qquad (16)$$

Substitution of Eq. 15 and Eq. 16 into Eq. 12 yields:

$$\Delta p = \frac{q\, e^{-\frac{r_m^2}{4\eta t}}}{2\phi c\eta t}\,\frac{h_f}{h}\left\{1 + \frac{2h}{\pi h_f}\sum_{n=1}^{\infty}\frac{1}{n}\exp\left(-\frac{n^2\pi^2\eta t}{h^2}\right)\left[\sin\frac{n\pi}{h}(z_f + \frac{h_f}{2})\right.\right.$$

$$\left.\left.- \sin\frac{n\pi}{h}(z_f - \frac{h_f}{2})\right]\cos\frac{n\pi z_m}{h}\right\}\int_0^{r_f} I_o\left(\frac{r_m r}{2\eta t}\right) e^{-\frac{r^2}{4\eta t}}\, r dr$$

which can be rearranged into:

$$\Delta p = \frac{e^{-\frac{r_m^2}{4\eta t}}}{2\phi c\eta t} \; \frac{h_f}{h} \left\{ 1 + \frac{4h}{\pi h_f} \sum_{n=1}^{\infty} \frac{1}{n} \exp\left(- \frac{n^2\pi^2\eta t}{h^2} \right) \sin \frac{n\pi h_f}{2h} \right.$$

$$\left. \cos \frac{n\pi z_f}{h} \; \cos \frac{n\pi z_m}{h} \right\} \int_0^{r_f} I_o \left(\frac{r_m r}{2\eta t} \right) e^{-\frac{r^2}{4\eta t}} \, rdr \qquad (17)$$

Eq. 17 represents the pressure drop created at M by a single, horizontal, instantaneous fracture of strength **q** per time unit, in an infinite reservoir with no flow through the upper and lower boundaries.

The pressure drop created by a continuous horizontal fracture source is then given by:

$$\Delta p = \frac{q}{\cdots} \int \left| 1 + \frac{4h}{\cdots} \sum_{n=1} e^{-\frac{n^2\pi^2\eta(t-t')}{h^2}} \sin \frac{n\pi h_f}{\cdots} \right.$$

$$\left. \cos n\pi \frac{z_f}{h} \; \cos n\pi \frac{z_m}{h} \right\| \int_0^{r_f} I_o \left(\frac{r_m r}{2\eta(t-t')} \right) e^{-\frac{r^2}{4\eta(t-t')}} \, rdr \Bigg]$$

$$\frac{e^{-\frac{r_m^2}{4\eta(t-t')}}}{t-t'} \, dt' \qquad (18)$$

Eq. 18 can be written in a more compact form by using $t'=t-t'$ in the integrand:

$$\Delta p = \frac{q}{2\phi c\eta}\frac{h_f}{h}\int_0^t \left[1 + \frac{4h}{\pi h_f}\sum_{n=1}^{\infty}\frac{1}{n}e^{-\frac{n^2\pi^2\eta t'}{h^2}}\sin\frac{n\pi h_f}{2h}\cos n\pi\frac{\ }{h}\right.$$

$$\left.\cos n\pi\frac{z_m}{h}\right]\left[\int_0^{r_f} I_o\left(\frac{r_m r}{2\eta t'}\right)e^{-\frac{r^2}{4\eta t'}}\, r dr\right]\frac{e^{-\frac{r_m^2}{4\eta t'}}}{t'}\, dt' \qquad (19)$$

Noting that the constant withdrawal rate from the fracture is given by:

$$q_f = \int_0^{r_f} r\, dr \int_{z_f - \frac{h_f}{2}}^{z_f + \frac{h_f}{2}} dz \int_0^{2\pi} q\, d\theta$$

or:

$$q_f = \pi r_f^2\, h_f\, q \qquad (20)$$

and recalling that the real vertical coordinate has been divided by $\sqrt{\frac{k_r}{k_z}}$ for deriving the solution, **Eq.** 19 can be rewritten as:

$$\Delta p = \frac{q_f}{2\pi r_f^2\, h\phi c\eta}\int_0^t \left[1 + \frac{4h}{\pi h_f}\sum_{n=1}^{\infty}\frac{1}{n}e^{-\frac{n^2\pi^2\eta k_z t'}{h^2 k_r}}\sin\frac{n\pi h_f}{2h}\cos n\pi\frac{z_f}{h}\right.$$

$$\left.\cos n\pi\frac{z_m}{h}\right]\left[\int_0^{r_f} I_o\left(\frac{r_m r}{2\eta t'}\right)e^{-\frac{r^2}{4\eta t'}}\, r dr\right]\frac{e^{-\frac{r_m^2}{4\eta t'}}}{t'}\, dt' \qquad (21)$$

Eq. 21 represents the solution to the problem we have defined at the beginning of this chapter.

We define the following dimensionless expressions:

$$p_D\,(r_D,\ z_D,\ t_D) = \frac{2\pi kh}{q_f\ \mu}\,\Delta p \tag{22}$$

$$t_D = \frac{k_r\ t}{\phi\mu c\ r_f^2} \tag{23}$$

$$r_D = \frac{r_m}{r_f} \tag{24}$$

$$z_D = \frac{z_m}{r_f}\sqrt{\frac{k_r}{k_z}} \tag{25}$$

From **Eq.** 25, we define the dimensionless reservoir thickness as $h_D = \frac{h}{r_f}\sqrt{\frac{k_r}{k_z}}$; replacing η by $\frac{k_r}{\phi\mu c}$, and substituting the dimensionless expressions defined by **Eqs.** 22 to 25 into **Eq.** 21, we obtain the dimensionless pressure drop at any point in the reservoir:

$$p_D(r_D,\ z_D,\ t_D) = \int_0^{t_D}\left[1 + \frac{4h}{\pi h_f}\sum_{n=1}^{\infty}\frac{1}{n}\,e^{-\frac{n^2\pi^2 t_D'}{h_D^2}}\ \sin n\pi\,\frac{h_f}{2h}\ \cos n\pi\,\frac{z_f}{h}\ \cos n\pi\,\frac{z}{h}\right]\left[\int_0^1 I_o\left(\frac{r_D r_D'}{2t_D'}\right) e^{-\frac{r_D'^2}{4t_D'}}\ r_D'\,dr_D'\right]\frac{e^{-\frac{r_D^2}{4t_D'}}}{t_D'}\,dt_D' \tag{26}$$

In Eq. 26, the ratios $\frac{h_f}{h}$, $\frac{z_f}{h}$, and $\frac{z}{h}$, already dimensionless, have not been modified; t'_D and r'_D refer to rhe dimensionless variables of integration; z_m has been replaced by z since there is no ambiguity in this symbolism. We notice that the integrand in Eq. 26 is the product of two functions, one depending on r_D and $t_{D'}$ the other depending on t_D and the vertical dimensions.

The expression $\frac{e^{-\frac{r_D^2}{4t_D}}}{t_D} \int_0^1 I_o\left(\frac{r_D r'_D}{2t_D}\right) e^{-\frac{r_D^{2'}}{4t_D}} r'_D \, dr'_D$ is known as the P function* and is commonly encountered in problems in applied mathematics involving surface integration in cylindrical coordinates. It has been studied and tabulated by J. I. Masters[26]. Using Master's notations, we write:

$$\frac{e^{-\frac{r_D^2}{4t_D}}}{t_D} \int_0^1 I_o\left(\frac{r_D r'_D}{2t_D}\right) e^{-\frac{r_D'^2}{4t_D}} r'_D \, dr'_D = 2P\left(\frac{1}{\sqrt{2t_D}}, \frac{r_D}{\sqrt{2t_D}}\right) \tag{27}$$

The other function, which we shall call the Z function, is also a common one in reservoir engineering problems and has appesred in several publications[27,28]. It is actually an unsteady Green's function for an infinite slab with impermeable boundaries[29, 30]. We define:

*This fact is mentioned by Carlaw and Jaeger[25], and has been brought to the attention of the author by R. Al-Hussainy.

$$Z\left(\frac{t_D}{h_D^2}, \frac{h_f}{2h}, \frac{z_f}{h}, \frac{z}{h}\right) =$$

$$1 + \frac{4h}{\pi h_f} \sum_{n=1}^{\infty} \frac{1}{n} \epsilon^{-\frac{n^2\pi^2 t_D}{h_T^2}} \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cdot \cos n\pi \frac{z}{h} \tag{28}$$

Eq. 26 thus becomes:

$$p_D\ (r_D,\ z_D,\ t_D) = 2 \int \left(\frac{1}{\sqrt{2t_D'}}, \frac{r_D}{\sqrt{2t_D'}}\right) Z \left(\frac{t_D'}{h_D^2}, \frac{h_f}{2h}, \frac{z_f}{h}, \frac{z}{h}\right) dt_D' \tag{29}$$

In order to make use of some other properties of the pressure drop function, we will also write Eq. 26 in the form:

$$p_D\ (r_D,\ z_D,\ t_D) = p_D\ (r_D,\ t_D) + \sigma\ (r_D,\ z_D,\ t_D) \tag{30}$$

where

$$p_D\ (r_D,\ t_D) = 2\int_0^{t_D} P\left(\frac{1}{\sqrt{2t_D'}}, \frac{r_D}{\sqrt{2t_D'}}\right) dt_D' \tag{31}$$

and

$$\sigma\ (r_D,\ z_D,\ t_D) = \frac{8h}{\pi r_f} \sum_{n=1}^{\infty} \frac{1}{n} \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h}$$

$$\int_0^{t_D} \exp\left(-\frac{n^2\pi^2 t_D'}{h_D^2}\right) P\left(\frac{1}{\sqrt{2t_D'}}, \frac{r_D}{\sqrt{2t_D'}}\right) dt. \tag{32}$$

or

$$\sigma\ (r_D,\ z_D,\ t_D) = 2 \int^{t} \left(\frac{1}{\sqrt{2t_D'}} \quad \frac{}{\sqrt{2t_D'}} \right| \left[Z\left(\frac{t_D'}{h_D^2},\ \frac{h_f}{2h},\ \frac{z_f}{h},\ \frac{z}{h} \right) - 1 \right] dt_D' \tag{33}$$

Eq. 31 represents the pressure drop created at any point in the reservoir by *a* solid cylinder source of radius r_f which completely penetrates the formation. Eq. 31 is direcrly derived in Appendix A, and it is shown that, if $r_f \to 0$, Eq. 31 reduces to the well-known line source solution[31].

By analogy with the skin factor defined by Van Everdingen[32] and Hurst[33], the σ function defined by Eq. 32 is called the pseudo-skin funcrion, The actual skin factor S as defined by Van Everdingen[32] and Hurst[33] was identified as a pressure drop across a damaged "skin" at the wellbore. We can extend their definition as follows:

$$s(r_D,\ z_D,\ t_D) = p_D\ (r_D, z_D,\ t_D) - \left[-\frac{1}{2}\, Ei \left(-\frac{r_D^2}{4t_D} \right) \right],\ r_D > 0 \tag{34}$$

where $-\frac{1}{2} E_i \left(-\frac{r_D^2}{4t_D} \right)$ represents the line source solution.

Eqs. 30 and 34 can be combined to yield:

$$s\ (r_D,\ z_D,\ t_D) = \frac{1}{2}\, Ei \left(-\frac{r_D^2}{4t_D} \right) + p_D\ (r_D,\ t_D) + \sigma\ (r_D,\ z_D,\ t_D),\ r_D > 0 \tag{35}$$

which shows that, if $r_D > 1$, the pseudo skin function becomes identical to the actual skin function for large values of time, or even for any value of t_D if $r_f \to 0$, as will be shown later.

In the following chapters, we will study the behavior of Eq. 26 at small and large values of time. Because Eq. 26 is governed by the behavior of the P and Z functions, we will first study these two functions.

B. PROPERTIES OF THE P FUNCTION

We will emphasize some properties of the P function that are of interest here. We use the Laplace transform of P to study its behavior at small values of t_D.

1. Laplace transform

The Laplace transform of $\frac{1}{2t} e^{-(x^2 + x'^2)/4Kt} I_\nu\left(\frac{xx'}{2Kt}\right)$, $\nu \geq 0$ is given by[34].

$$\begin{cases} I_\nu(x'\sqrt{s})\, K_\nu\, (x\sqrt{s}), & x > x' \\ I_\nu(x\sqrt{s})\, K_\nu\, (x'\sqrt{s}), & x < x' \end{cases}$$

a. $\underline{r_D > 1}$ the Laplace transform of the P function is:

$$L\{P\} = K_o\, (r_D\sqrt{s}) \int_0^1 I_o\, (r'_D\sqrt{s})\, r'_D\, dr'_D \tag{36}$$

The integral can be evaluated by using the properties of Bessel functions. If B(u) is a Bessel function of zero order, it is a solution of the differential equation[35].

$$uB''(u) + B'(u) - uB(u) = 0$$

and then:

$$\int u\, B\,(u)\, du = \int u\, B''\,(u)\, du + \int B'\,(u)\, du$$

Integrating by parts, we can show that:

$$\int u\, B''\,(u)\, du = u\, B'\,(u) - \int B'\,(u)\, du$$

and therefore:

$$\int u\, B\,(u)\, du = u\, B'\,(u) \tag{37}$$

Hence, Eq. 36 leads to:

$$L\,\{P\} = \frac{K_o(r_D\sqrt{s})\; I_1(\sqrt{s})}{\sqrt{s}} \tag{38}$$

b. $\underline{0 < r_D < 1}$

The Laplace transform of the P function is given by[34]:

$$L\,\{P\} = K_o(r_D\sqrt{s}) \int_0^{r_D} I_o\,(r_D'\sqrt{s})\; r_D'\; dr_D'$$

$$+ I_o\,(r_D\sqrt{s}) \int_{r_D}^{1} K_o\,(r_D'\sqrt{s})\; r_D'\; dr_D' \tag{39}$$

From Eq. 37, this can be simplified to yield:

$$L\{P\} = \frac{K_o(r_D\sqrt{s})\ r_D\sqrt{s}\ I_1\ (r_D\sqrt{s})}{s} - \frac{I_o\ (r_D\sqrt{s})\sqrt{s}\ K_1\ (\sqrt{s})}{s}$$

$$+ \frac{I_o\ (r_D\sqrt{s})\ r_D\sqrt{s}\ K_1\ (r_D\sqrt{s})}{s} \qquad (40)$$

or

$$L\{P\} = \frac{r_D}{\sqrt{s}}\left[K_o\ (r_D\sqrt{s})\ I_1\ (r_D\sqrt{s}) + K_1\ (r_D\sqrt{s})\ I_o\ (r_D\sqrt{s})\right]$$

$$- \frac{I_o\ (r_D\sqrt{s})\ K_1\ (\sqrt{s})}{\sqrt{s}} \qquad (41)$$

The term between brackets is known[35] to be equal to $\frac{1}{r_D\sqrt{s}}$ and we finally obtain:

$$L\{P\} = \frac{1}{s} - \frac{I_o\ (r_D\sqrt{s})\ K_1\ (\sqrt{s})}{\sqrt{s}} \qquad (42)$$

c. $\underline{r_D = 1}$

From Eq. **38** or Eq. *42* in which $r_D = 1$, we obtain:

$$L\{P\} = \frac{K_o\ (\sqrt{s})\ I_1\ (\sqrt{s})}{\sqrt{s}} \qquad (43)$$

d. $\underline{r_D = 0}$

An analytical expression for $r_D = 0$ is obtained directly from the definition of the P function:

$$P\left(\frac{1}{\sqrt{2t_D}}, 0\right) = \frac{1}{2t_D}\int_0^{1} e^{-\frac{r_D'^2}{4t_D}}\ r_D'\ dr_D' = \left(1 - e^{-\frac{1}{4t_D}}\right) \qquad (44)$$

2. <u>Short-time behavior</u>

Since s and t_D are inversely related, $s \to \infty$ as $t_D \to 0$. Using asymptotic forms for the Bessel functions, we obtain at early times[36], as $s \to \infty$:

$$K_\nu (r_D\sqrt{s}) \sim \frac{e^{-r_D\sqrt{s}}}{\sqrt{\frac{2r_D}{\pi}\sqrt{s}}} \quad , \quad \nu = 0, 1$$

$$I_\nu (r_D\sqrt{s}) \sim \frac{e^{r_D\sqrt{s}}}{\sqrt{2\pi r_D\sqrt{s}}} \quad , \quad \nu = 0, 1$$

We then obtain the following limits for the P function:

a. <u>$r_D > 1$</u>

as $s \to \infty$:

$$L\{P\} \sim \frac{e^{-(r_D - 1)\sqrt{s}}}{2s\sqrt{r_D}} \tag{45}$$

and inversion of Eq. 45 yields[31]:

$$P \sim \frac{1}{2\sqrt{r_D}} \quad \mathrm{erfc}\, \frac{r_D - 1}{2\sqrt{t_D}} \quad \text{as } t_D \to 0 \tag{46}$$

and P approaches zero as time approaches zero.

$P\left(\frac{1}{\sqrt{2t_D}}, \frac{r_D}{\sqrt{2t_D}}\right)$ is less than 0.01 when:

$$t_D \leqslant \frac{(r_D - 1)^2}{20} \tag{47}$$

b. $\underline{0 < r_D < 1}$

as $s \to \infty$:

$$L\{P\} \sim \frac{1}{s} - \frac{e^{-(1 - r_D)\sqrt{s}}}{2s\sqrt{r_D}} \tag{48}$$

The Laplace inversion gives34.

$$P \sim 1 - \frac{1}{2\sqrt{r_D}} \text{ erfc } \frac{1 - r}{2\sqrt{t_D}} \quad \text{as } t_D \to 0 \tag{49}$$

and P approaches unity as time approaches zero.

$P\left(\frac{1}{\sqrt{2t_D}}, \frac{r_D}{\sqrt{2t_D}}\right)$ differs from 1 by less than 1% when:

$$t_D \leqslant \frac{(1 - r_D)^2}{20} \tag{50}$$

c. $\underline{r_D = 1}$

From Eq. 45 and Eq. 48, it follows that, as $s \to \infty$:

$$L\{P\} \sim \frac{1}{2s} \tag{51}$$

and the inversion of Eq. 51 yields:

$$\lim_{t_D \to 0} P = \frac{1}{2} \tag{52}$$

Thus P approaches $\frac{1}{2}$ as time approaches zero.

Eq. 52 does not provide a value of t_D below which P can be considered as constant. This can be done by deriving an analytical expression for $P\left(\frac{1}{\sqrt{2t_D}}, \frac{1}{\sqrt{2t_D}}\right)$ which is possible for $r_D = 1$. It can be shown[37] that:

$$\int_0^1 I_o\left(\frac{r'_D}{2t_D}\right) e^{-\frac{r'^2_D}{4t_D}} r'_D \, dr'_D = t_D \, e^{\frac{1}{4t_D}} \left[1 - e^{-\frac{1}{2t_D}} I_o\left(\frac{1}{2t_D}\right)\right]$$

therefore:

$$P\left(\frac{1}{\sqrt{2t_D}}, \frac{1}{\sqrt{2t_D}}\right) = \frac{1}{2}\left[1 - e^{-\frac{1}{2t_D}} I_o\left(\frac{1}{2t_D}\right)\right] \tag{53}$$

When $t_D \to 0$, Eq. 53 yfelds:

$$P\left(\frac{1}{\sqrt{2t_D}}, \frac{1}{\sqrt{2t_D}}\right) \sim \frac{1}{2}\left(1 - \sqrt{\frac{t_D}{\pi}}\right) \tag{54}$$

and P differs from $\frac{1}{2}$ by less than 1% when:

$$t_D \leqslant 10^{-4}\,\pi \tag{55}$$

d. $\underline{r_D - 0}$

From Eq. 44:

$$\lim_{t_D \to 0} P\left(\frac{1}{\sqrt{2t_D}}, 0\right) = 1$$

and P differs from 1 by less than 1% when $t_D \leqslant \frac{1}{20}$.

It thus appears that each time limit is function of the distance from the point to r_f, $|r - r_f|$, except for $r_D = 1$ where the time limit depends upon r_f.

3. <u>Long-time behavior</u>

The P function can be expanded in Taylor's series as t_D increases. Using expansions similar to those given in Appendix A by Eqs. A-7 and A-8 for the Bessel function and the exponentials, we obtain:

$$P\left(\frac{1}{\sqrt{2t_D}}, \frac{r_D}{\sqrt{2t_D}}\right) = \frac{1}{4t_D} - \frac{2r_D^2 + 1}{32t_D^2} - O\left(\frac{1}{t_D^3}\right) \tag{56}$$

and the P function is equivalent to $\frac{1}{4t_D}$ within 1% as soon as:

$$t_D \geqslant 12.5\ (2r_D^2 + 1) \tag{57}$$

C. <u>PROPERTIES OF THE Z FUNCTION</u>

1. <u>Laplace transform and short time behavior.</u>

Here again, it is convenient to use the Laplace transform of Z to study the behavior of the function at small values of time.

We define:

$$L\{Z\} = \int_0^{\infty} e^{-st_D}\, Z\left(\frac{t_D}{h_D^2}, \frac{h_f}{2h}, \frac{z_f}{h}, \frac{z}{h}\right) dt_D \tag{58}$$

and the Z function is given by Eq. 28. Thus:

$$L\{Z\} = \frac{1}{s}\left[1 + \frac{4h}{\pi h_f}\sum_{n=1}^{\infty}\left(\frac{1}{n} - \frac{n}{n^2 + \frac{h_D^2}{\pi^2}s}\right)\sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h}\right] \tag{59}$$

The product of trigonometric functions can be written:

$$\begin{aligned} 4 \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h} &= \sin n\pi \frac{\frac{h_f}{2} - (z_f + z)}{h} \\ &+ \sin n\pi \frac{\frac{h_f}{2} + z_f + z}{h} + \sin n\pi \frac{\frac{h_f}{2} - z_f + z}{h} \\ &+ \sin n\pi \frac{\frac{h_f}{2} + z_f - z}{h} \end{aligned} \tag{60}$$

and it is known[38] that, if $0 < x < 2\pi$:

$$\sum_{k=1}^{\infty} \frac{\sin kx}{k} = \frac{\pi - x}{2} \tag{61}$$

$$\sum_{k=1}^{\infty} \frac{k \sin kx}{k^2 + \alpha^2} = \frac{\pi}{2} \frac{\sinh \alpha(\pi - x)}{\sinh \alpha\pi} \tag{62}$$

In order to apply **Eqs.** 61 and 62, we must consider different cases depending upon z.

a. $\underline{z_f + \frac{h_f}{2} < z \leqslant h}$

The point is above the fracture; we thus write **Eq.** 60 as:

$$4 \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h} = - \sin n\pi \frac{z + (z_f - \frac{h_f}{2})}{h}$$

$$+ \sin n\pi \frac{z \bullet (z_f + \frac{h_f}{2})}{h} + \sin n\pi \frac{z - (z_f - \frac{h_f}{2})}{h}$$

$$- \sin n\pi \frac{z - (z_f + \frac{h_f}{2})}{h} \tag{63}$$

where all arguments on the right hand side are between 0 and 2π. Then, use of **Eq.** 61 leads to:

$$\sum_{n=1}^{\infty} \frac{1}{n} \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h} = - \frac{\pi h_f}{4h} \tag{64}$$

and Eq. 62 yields:

$$\sum_{n=1}^{\infty} \frac{n}{n^2 + \frac{h_D^2}{\ } s} \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h} = \frac{\pi U}{8 \sinh h_D \sqrt{s}} \qquad (65)$$

where

$$U = -\sinh \left[h - z - (z_f - \frac{h_f}{2})\right]_D \sqrt{s} + \sinh \left[h - z - (z_f + \frac{h_f}{2})\right]_D \sqrt{s}$$

$$+ \sinh \left[h - z + (z_f - \frac{h_f}{2})\right]_D \sqrt{s} - \sinh \left[h - z + (z_f + \frac{h_f}{2})\right]_D \sqrt{s} \qquad (66)$$

Eq. 66 can be simplified to:

$$U = -4 \cosh (h - z)_D \sqrt{s} \cosh z_{f_D} \sqrt{s} \sinh \frac{h_{f_D}}{2} \sqrt{s} \qquad (67)$$

Substitution of Eqs. 64 and 65 into Eq. 59 yields:

$$L\{Z\} = \frac{2h}{h_f} \frac{\sinh \frac{h_{f_D}}{2} \sqrt{s} \cosh z_{f_D} \sqrt{s} \cosh (h - z)_D \sqrt{s}}{s \sinh h_D \sqrt{s}} \qquad (68)$$

Since t_D and s are inversely related, the behavior of Z at early times is given by taking the limit of L {Z} as $s \rightarrow \infty$. Eq. 68 thus yields as $s \rightarrow \infty$:

$$L\{Z\} \sim \frac{h}{h_f} \frac{e^{-\left[z - (z_f + \frac{h_f}{2})\right]_D \sqrt{s}}}{2s} \qquad (69)$$

The form of Z at early time is obtained by inverting Eq. 69, which yields[34].

$$Z \sim \frac{h}{2h_f} \operatorname{erfc} \frac{\left[z - \left(z_f + \frac{h_f}{2}\right)\right]_D}{2\sqrt{t_D}} \quad \text{as } t_D \to 0 \tag{70}$$

As t_D decreases, Z decreases and becomes negligible when:

$$t_D \leqslant \frac{\left[z - \left(z_f + \frac{h_f}{2}\right)\right]_D^2}{20} \tag{71}$$

which can also be expressed in terms of real variables as:

$$\frac{k_z\ t}{\phi\mu c \left[z - \left(z_f + \frac{h_f}{2}\right)\right]^2} \leqslant \frac{1}{20} \tag{72}$$

b. $\underline{0 \leqslant z < z_f - \frac{h_f}{2}}$

The point is below the fracture. Eq. 60 becomes:

$$4 \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h} = - \sin n\pi \frac{\left(z_f - \frac{h_f}{2}\right) + z}{h}$$

$$+ \sin n\pi \frac{\left(z + \frac{h_f}{2}\right) + z}{h} - \sin n\pi \frac{\left(z_f - \frac{h_f}{2}\right) - z}{h}$$

$$+ \sin n\pi \frac{\left(z_f + \frac{h_f}{2}\right) - z}{h} \tag{73}$$

The same calculations as before yield:

$$L\{Z\} = \frac{2h}{h_f} \sinh \frac{h_{fD}}{2} \sqrt{s} \; \frac{\cosh (h - z_f)_D \sqrt{s} - \cosh z_D \sqrt{s}}{s \sinh h_D \sqrt{s}} \tag{74}$$

from which it follows that:

$$L\{Z\} \sim \frac{h}{h_f} \; \frac{e^{-\left[(z_f - \frac{h_f}{2}) - z\right]_D \sqrt{s}}}{} \quad \text{as } s \to \infty \tag{75}$$

The inversion of Eq. 75 yields[34]:

$$Z \sim \frac{h}{h_f} \; \text{erfc} \; \frac{\left[(z_f - \frac{h_f}{2}) - z\right]_D}{2\sqrt{t_D}} \quad -- \; t_D \to \tag{76}$$

and Z is negligible when:

$$t_D \leqslant \frac{\left[(z_f - \frac{h_f}{2}) - z\right]_D^2}{20} \tag{77}$$

or

$$\frac{k_z \, t}{\phi \mu c \left[(z_f - \frac{h_f}{2}) - z\right]^2} \leqslant \frac{1}{20} \tag{78}$$

c. $z_f - \frac{h_f}{2} \leqslant z \leqslant z_f + \frac{h_f}{2}$

The point is inside the fracture; Eq. 58 must be written as:

$$4 \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h} = - \sin n\pi \frac{z + (z_f - \frac{h_f}{2})}{h}$$

$$+ \sin n\pi \frac{(z_f + \frac{h_f}{2}) + z}{h} + \sin n\pi \frac{z - (z_f - \frac{h_f}{2})}{h}$$

$$+ \sin n\pi \frac{(z_f + \frac{h_f}{2}) - z}{h} \qquad (79)$$

so that all arguments are between 0 and 2π.

From Eq. 61, we obtain:

$$\sum_{n=1}^{\infty} \frac{1}{n} \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{\ } \cos n\pi \frac{z}{h} = \frac{\pi}{4} - \frac{\pi h_f}{4h} \qquad (80)$$

and from Eq. 62:

$$\sum_{n=1}^{\infty} \frac{r}{n^2 + \frac{h_D^2}{\ } s} \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{\ }{h} \cos n\pi \frac{\ }{h} = \frac{\pi V}{8 \sinh h_D \sqrt{s}} \qquad (81)$$

where

$$V = - \sinh \left[h - z - (z_f - \frac{h_f}{2})\right]_D \sqrt{s} + \sinh \left[h - z - (z_f + \frac{h_f}{2})\right]_D \sqrt{s}$$

$$+ \sinh \left[h - z + (z_f - \frac{h_f}{2})\right]_D \sqrt{s} + \sinh \left[h + z - (z_f + \frac{h_f}{2})\right]_D \sqrt{s}$$

$$(82)$$

and Eq. 59 can be written as:

$$L\{Z\} = \frac{h}{h_f}\,\frac{1}{s}\left(1 - \frac{V}{2\sinh h_D\sqrt{s}}\right) \tag{83}$$

As t_D decreases, or s increases, Eq. 83 yields:

$$L\{Z\} \sim \frac{1}{s}\left\{1 + \frac{1}{2}\left[e^{-(z_f - \frac{h_f}{2} + z)_D\sqrt{s}} - e^{-(z_f + \frac{h_f}{2} + z)_D\sqrt{s}} - e^{-(z - z + \frac{h_f}{2})_D\sqrt{s}} - e^{-(z_f + \frac{h_f}{2} - z)_D\sqrt{s}}\right]\right\} \tag{84}$$

Eq. 84 is thus equivalent to:

$$L\{Z\} \sim \frac{h}{h_f}\,\frac{1}{s}\left(1 - \frac{1}{2}e^{-[(z_f + \frac{h_f}{2}) - z]_D\sqrt{s}}\right), \text{ if } z_f \leqslant z < z_f + \frac{h_f}{2} \tag{85}$$

$$L\{Z\} \sim \frac{h}{h_f}\,\frac{1}{s}\left(1 - \frac{1}{2}e^{-[z - (z_f - \frac{h_f}{2})]_D\sqrt{s}}\right), \text{ if } z_f - \frac{h_f}{2} < z \leqslant z_f \tag{86}$$

Inverting Eqs. 85 and 86 yields, as $t_D \to 0$:

$$Z \sim \frac{h}{h_f} - \frac{h}{2h_f}\,\mathrm{erfc}\,\frac{[(z_f + \frac{h_f}{2}) - z]_D}{2\sqrt{t_D}}, \text{ if } z_f \leqslant z < z_f + \frac{h_f}{2} \tag{87}$$

$$Z \sim \frac{h}{h_f} - \frac{h}{2h_f}\,\mathrm{erfc}\,\frac{[z - (z_f - \frac{h_f}{2})]_D}{2\sqrt{t_D}}, \text{ if } z_f - \frac{h_f}{2} < z \leqslant z_f \tag{88}$$

It thus follows that:

$$\lim_{t_D \to 0} Z = \frac{h}{h_f} \tag{89}$$

Z is practically equal to $\frac{h}{h_f}$ as soon as:

$$\begin{cases} t_D \leqslant \dfrac{[(z_f + \frac{h_f}{2}) - z]}{20} & \text{, for Eq. 85} \\[2ex] t_D \leqslant \dfrac{[z - (z_f + \frac{h_f}{2})]}{20} & \text{, for Eq. 86} \end{cases} \tag{90}$$

If we take $z = z_f + \varepsilon x \frac{h_f}{2}$, where $0 \leqslant x < 1$ and $\varepsilon = \pm 1$, the time limits given in Eq. 90 become:

$$\frac{k_z t}{\phi \mu c (1 - x)^2 h_f^2} \leqslant \frac{1}{80} \tag{91}$$

which depends only upon h_f, the altitude of the point in the fracture, and the vertical permeability.

When $z = z_f + \frac{h_f}{2}$ or $z = z_f - \frac{h_f}{2}$, $x = 1$, and the time at which Z reaches its early time constant value, must be derived directly from Eq. 83, which then can be written as:

$$L\,\{Z\} = \frac{h}{h_f}\;\frac{1}{s}$$

$$\left(1 - \frac{-\sinh\left[h - 2z_f + (1-\varepsilon)\frac{h_f}{2}\right]_D\sqrt{s} + \sinh\left[h - 2z_f - (1+\varepsilon)\frac{h_f}{2}\right]_D\sqrt{s}}{2\sinh h_D\sqrt{s}}\right.$$

$$\left. - \frac{\sinh\left[h - (1+\varepsilon)\frac{h_f}{2}\right]_D\sqrt{s} + \sinh\left[h - (1-\varepsilon)\frac{h_f}{2}\right]_D\sqrt{s}}{2\sinh h_D\sqrt{s}}\right) \tag{92}$$

and as $s \to \infty$:

$$L\,\{Z\} \sim \frac{h}{h_f}\;\frac{1}{s}\left\{1 - \frac{1}{2}\left[- e^{-\left[2z_f - (1-\varepsilon)\frac{h_f}{2}\right]_D\sqrt{s}} + e^{-\left[2z_f + (1+\varepsilon)\frac{h_f}{2}\right]_D\sqrt{s}} + e^{-(1+\varepsilon)\frac{h_{f_D}}{2}\sqrt{s}} + e^{-(1-\varepsilon)\frac{h_{f_D}}{2}\sqrt{s}}\right]\right\} \tag{93}$$

Eq. 93 leads to:

$$L\,\{Z\} \sim \frac{h}{2h_f}\,\frac{1}{s}\left(1 - e^{-h_{f_D}\sqrt{s}}\right) \text{as } s \to \infty \tag{94}$$

and therefore, as $t_D \to 0$:

$$Z \sim \frac{h}{2h_f} - \frac{h}{2h_f}\,\mathrm{erfc}\,\frac{h_{f_D}}{2\sqrt{t_D}} \tag{95}$$

and Z can be taken equal to $\frac{h}{2h_f}$ as soon as:

$$t_D \leqslant \frac{h_{f_D}^2}{20} \tag{96}$$

or

$$\frac{k_z t}{\phi \mu c\, h_f^2} \leqslant \frac{1}{20} \tag{97}$$

2. Long-time behavior

As t_D increases, the term $e^{-\frac{n^2 \pi^2 t_D}{h_D^2}}$ in Eq. 28 decreases rapidly, and the number of terms in the infinite sum which is required to reach a given accuracy decreases, until the first term itself can be neglected.

Therefore, the Z function will be different from unity by less than 1% as soon as:

$$t_D \geqslant \frac{5 h_D^2}{\pi^2} \tag{98}$$

or

$$\frac{k_z t}{\phi \mu c\, h^2} \geqslant \frac{5}{\pi^2} \tag{99}$$

D. LONG TIME PRESSURE BEHAVIOR

In well test analysis, the pressure drop at the well is generally plotted versus the logarithm of time. In the case of a slightly compressible fluid, a straight line of slope $\frac{1}{2}$ is obtained after a certain time, which defines the beginning of the transient flow period[39].

If we make such a plot in the case of a horizontal fracture, we obtain a curve the slope of which varies with time, and is given by:

$$\frac{dp_D\ (r_D,\ z_D,\ t_D)}{d\ (\ell n\ t_D)} = 2\ t_D\ P\left(\frac{1}{\sqrt{2t_D}}\ ,\ \frac{r_D}{\sqrt{2t_D}}\right) Z\left(\frac{t_D}{h_D^2}\ ,\ \frac{h_f}{2h}\ ,\ \frac{z_f}{h}\ ,\ \frac{z}{h}\right) \tag{100}$$

In the preceding sections, we have shown that at long times, the P function approaches $(1/4\ t_D)$ and the Z function approaches unity. Thus the value of the slope will differ from $\frac{1}{2}$ by less than 1% as soon as **Eq. 57** and **Eq.** 98 are satisfied. That is, as soon as:

$$t_D \geqslant 12.5\ (2r_D^2 + 1) \tag{57}$$

and

$$t_D \geqslant \frac{5h_D^2}{\pi^2} \tag{98}$$

It is also interesting to inspect long-time pressure behavior by means of **Eq**. 30:

$$p_D\ (r_D,\ z_D,\ t_D) = p_D\ (r_D,\ t_D) + \sigma(r_D,\ z_D,\ t_D) \tag{30}$$

where the pseudo-skin function σ is given by Eq. 33:

$$\sigma(r_D,\ z_D,\ t_D) = 2\int_0^{t_D} P\left(\frac{1}{\sqrt{2t'_D}}\ ,\ \frac{r_D}{\sqrt{2t'_D}}\right)$$

$$\left[Z\left(\frac{t'_D}{h_D^2}\ ,\ \frac{h_f}{2h}\ ,\ \frac{z_f}{h}\ ,\ \frac{z}{h}\right) - 1\right] dt \tag{33}$$

As time increases, the Z 'function approaches unity, the integrand in Eq. 33 approaches zero, and the σ function becomes constant. This constant, defined as the pseudo-skin factor is the maximum value of the pseudo-skin function and is equal to $\sigma(r_D, z_D, \frac{5h_D^2}{\pi^2})$. Using the results from Eqs. A-17 and A-18 in Appendix A, Eq. 30 can be written as:

$$p_D(r_D, z_D, t_D) = \frac{1}{2}\left[\ln \frac{t_D}{r_D^2} + 0.80907 + 2\sigma\ (r_D, z_D, \frac{5h_D^2}{\pi^2})\right], \text{ if } r_D \geq 1 \tag{101}$$

$$p_D(r_D, z_D, t_D) = \frac{1}{2}\left[\ln t_D + 1.80907 - r_D^2 + 2\sigma\ (r_D, z_D, \frac{5h_D^2}{\pi^2})\right], \text{ if } 0 \leq r_D \leq 1 \tag{102}$$

Eq. 101 shows that the pseudo skin factor is related to the skin factor S defined by Van Everdingen[32] and Hurst[33] for $r_D \geq 1$ and large values of t_D.

When $0 \leq r_D \leq 1$, we can express the pressure drop p_D as a function of the dimensionless time commonly used in the petroleum literature, $\frac{k_r t}{\phi c r_w^2}$, by writing:

$$p_D(r_D, z_D, t_D) = \frac{1}{2}\left\{\ln \frac{kt}{\phi\mu c\ r_w^2} + 0.80907 + 2\ \frac{1 - r_D^2}{2} - \ln \frac{r_f}{r_w} + \sigma(r_D, z_D, \frac{5h_D^2}{\pi^2})\right\} \tag{103}$$

E. WELLBORE PRESSURE

Because we are mainly interested in knowing the pressure at the wellbore, we should replace r_D by $\frac{r_w}{r_f}$ in Eq. 103. We thus obtain:

$$p_{wD} = \frac{1}{2}\left\{ \ln \frac{kt}{\phi \mu c\, r_w^2} + 0.80907 + 2\left[\frac{1 - r_{wD}^2}{2} + \ln r_{wD} + \sigma\left(r_{wD}, z_D, \frac{5h_D^2}{\pi^2}\right) \right] \right\} \tag{104}$$

We can write:

$$p_D(0, z_D, t_D) - p_{wD} = \frac{r_{wD}^2}{2} + \sigma\left(0, z_D, \frac{5h_D^2}{\pi^2}\right) - \sigma\left(r_{wD}, z_D, \frac{5h_D^2}{\pi^2}\right) \tag{105}$$

where:

$$\sigma\left(0, z_D, \frac{5h_D^2}{\pi^2}\right) - \sigma\left(r_{wD}, z_D, \frac{5h_D^2}{\pi^2}\right) = 2\int_0^{\frac{5h_D^2}{\pi^2}} \left[P\left(\frac{1}{\sqrt{2t'_D}}, 0\right) - P\left(\frac{1}{\sqrt{2t'_D}}, \frac{r_{wD}}{\sqrt{2t'_D}}\right)\right]\left[Z\left(\frac{t'_D}{h_D^2}, \frac{h_f}{2h}, \frac{z_f}{h}, \frac{z}{h}\right) - 1\right] dt'_D \tag{106}$$

From Eqs. 105 and 106, it can be concluded that the pressure at the wellbore differs from the pressure at the center of the fracture by less than 1% if:

$$r_w \leqslant \frac{r_f}{10} \tag{107}$$

for large values of time. This result is also valid at early times, from Eq. 50. We can therefore write, for all times:

$$p_{wD} = p_D\ (0,\ z_D,\ t_D) = 2\int_0^{t_D} P\left(\frac{1}{\sqrt{2t'_D}}\ ,\ 0\right)$$

$$Z\left(\frac{t'_D}{h_D^2}\ ,\ \frac{h_f}{2h}\ ,\ \frac{z_f}{h}\ ,\ \frac{z}{h}\right) dt'_D \tag{108}$$

which can simplified by means of Eq. *44* to:

$$p_{wD} = 2\int_0^{t_D}\left(1 - e^{-\frac{1}{4t'_D}}\right) Z\left(\frac{t'_D}{h_D^2}\ ,\ \frac{h_f}{2h}\ ,\ \frac{z_f}{h}\ ,\ \frac{z}{h}\right) dt'_D \tag{109}$$

or

$$p_{wD} = 2\int_0^{t_D}\left(1 - e^{-\frac{1}{4t'_D}}\right) dt'_D + \sigma(0,\ z_D,\ t_D)$$

which yields:

$$p_{wD} = -\frac{1}{2}\,Ei\left(-\frac{1}{4t_D}\right) + 2t_D\left(1 - e^{-\frac{1}{4t_D}}\right) + \sigma(0,\ z_D,\ t_D) \tag{110}$$

where

$$\sigma(0,\ z_D,\ t_D) = \frac{8h}{\pi h_f} \sum_{n=1}^{\infty} \frac{1}{n} \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h}$$

$$\int_0^{t_D} \left[1 - e^{-\frac{1}{4t'_D}} \right] e^{-\frac{n^2\pi^2 t'_D}{h_D^2}}\, dt'_D \tag{111}$$

The integral in Eq. **111** can be expressed as:

$$\int_0^{t_D} \left(1 - e^{-\frac{1}{4t'_D}}\right) e^{-\frac{n^2\pi^2 t'_D}{h_D^2}}\, dt'_D = \frac{h_D^2}{n^2\pi^2}\left(1 - e^{-\frac{n^2\pi^2 t_D}{h_D^2}}\right)$$

$$- \int_0^{t_D} e^{-\left(\frac{1}{4t'_D} + \frac{n^2\pi^2}{h_D^2} t'_D\right)}\, dt'_D \tag{112}$$

and it can be shown[40] that:

$$\int_0^{t_D} e^{-\left(\frac{1}{4t'_D} + \frac{n^2\pi^2}{h_D^2} t'_D\right)}\, dt'_D = \frac{h_D}{n\pi}\ K_1\left(\frac{n\pi}{h_D}\right)$$

$$- \int_0^{\frac{1}{4t_D}} e^{-\left[u + \frac{\left(\frac{n\pi}{h_D}\right)^2}{4u}\right]} \frac{du}{u^2} \tag{113}$$

Eq. 111 then becomes:

$$\sigma(0,\, z_D,\, t_D) = \frac{8\, h_D^3}{\pi\, h_{f_D}} \sum_{n=1}^{\infty} \frac{1}{n^3} \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h} \exp\left(- \frac{n^2 \pi^2 t_D}{h_D^2} \right)$$

$$- \frac{8\, h_D^2}{\pi^2\, h_{f_D}} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h} K_1\left(\frac{n\pi}{h_D}\right)$$

$$+ \frac{8\, h_D}{\pi\, h_{f_D}} \sum_{n=1}^{\infty} \frac{1}{n} \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h}$$

$$\int_0^{\frac{1}{4t_D}} e^{-\left[u + \frac{\left(\frac{n\pi}{h_D}\right)^2}{4u}\right]} \frac{du}{u^2} \tag{114}$$

and as t_D increases, **Eq.** 114 reduces **to:**

$$\sigma\left(0,\, z_D,\, \frac{5h_D^2}{\pi^2}\right) = \frac{8\, h_D^3}{\pi^3\, h_{f_D}} \sum_{n=1}^{\infty} \frac{1}{n^3} \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h}$$

$$- \frac{8\, h_D^2}{\pi^2\, h_{f_D}} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h} K_1\left(\frac{n\pi}{h_D}\right) \tag{115}$$

where $\sigma(0, z_D, \frac{5h_D^2}{\pi^2})$ is the pseudo-skin factor at $r_D = 0$.

Eq. 115 can be simplified using **Eq.** 79 and the following formula[38]:

$$\sum_{k=1}^{\infty} \frac{\sin kx}{k^3} = \frac{\pi^2 x}{6} - \frac{\pi x^2}{4} + \frac{x^3}{12}, \quad 0 \leqslant x \leqslant 2\pi$$

and we finally obtain the pseudo skin σ at the wellbore, for $z_f - \frac{h_f}{2} \leqslant z \leqslant z_f + \frac{h_f}{2}$:

$$\sigma = \frac{2}{3} h_D^2 - \frac{h_D (z - z_f)_D^2}{h_{f_D}} - h_D (z + z_f)_D - \frac{1}{4} h_D h_{f_D} + \frac{1}{12} h_{f_D}^2 + z_D^2 + z_{f_D}^2$$

$$- \frac{h_D^2}{\pi^2 h_{f_D}} \sum_{n=1}^{\infty} \frac{1}{n} \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h} K_1\left(\frac{n\pi}{h_D}\right) \quad (116)$$

Eq. 116 represents the constant pseudo-skin factor as a function of h_D, h_{f_D}, z_D and z_{f_D}. It is interesting to notice that, because the Bessel function K_1 decreases rapidly when the argument increases, it is not necessary to consider the infinite sum in calculations for $h_D \leqslant 1$.

F. RADIUS OF INFLUENCE

It is possible to find a value of r_D such that the constant pseudo skin factor $\sigma(r_D, z_D, \frac{5h_D^2}{\pi^2})$ be zero for greater values of r_D: we will call it the radius of influence of the fracture; at points beyond the radius of influence, the pseudo-skin factor due to the fracture is essentially zero, and the effect of the fracture has disappeared: the pressure drop is the same as created by the solid cylinder source of radius r_f. As pointed out in Appendix A, if r_D is sufficiently larger than 1, the pressure drop created by the solid cylinder source of radius r_f is equal to the pressure drop created by a line source well.

We have shown that for $r_D > 1$, $P\left(\frac{1}{\sqrt{2t_D}}, \frac{r_D}{\sqrt{2t_D}}\right) \leqslant \frac{1}{100}$ when $t_D \leqslant \frac{(r_D - 1)^2}{20}$. For a given t_D, the same relation holds when $r_D - 1 \geqslant 2\sqrt{5t_D}$. Therefore, if:

$$r_D - 1 \geqslant \frac{10}{\pi} h_D,$$

the pseudo skin factor $\sigma(r_D, z_D, \frac{5h_D^2}{\pi})$ is zero. Because $\sigma(r_D, z_D, \frac{5h_D^2}{\pi^2})$ is the maximum of the pseudo-skin function which is always positive, the pseudo-skin function itself is zero, and the pressure drop at all times is the same as that created by a solid cylinder.

The radius of influence of the fracture is therefore defined by:

$$r_{iD} - 1 = \frac{10}{\pi} h_D \tag{117}$$

This relation can be rewritten as:

$$r_i - r_f = \frac{10}{\pi} h \sqrt{\frac{k_r}{k_z}} \tag{118}$$

The horizontal. distance from the outer face of the fracture to rhe non perturbed zone is only a function of the thickness h of the reservoir, and of the permeability ratio $\frac{k_r}{k_z}$. It is independent of the fracture thickness h_f, and of the fracture radius r_f.

G. SHORT TIME PRESSURE BEHAVIOR

We have seen, in a preceding section of the present study that, if $0 \leqslant r_D - 1$, the P function can be approximated by 1 with less than 1% error if Eq. 50 is satisfied:

$$t_D \leqslant \frac{(1 - r_D)^2}{20} \tag{50}$$

or, in terms of real variables, if:

$$\frac{k_r t}{\phi \mu c (r_f - r)^2} \leqslant \frac{1}{20}$$

This time limit is obviously a function of the distance between the point and the radial boundary of the fracture.

When Eq. 50 is satisfied, we are left with a form of the pressure drop that is independent of the radial coordinate:

$$p_D (0 \leqslant r_D < 1, z_D, t_D) = 2 \int^{t} Z\left(\frac{t'_D}{h^2_-}, \frac{h_f}{2h}, \frac{z_f}{h}, \frac{z}{h}\right) dt'_D \tag{119}$$

and the behavior of p_D given by Eq. 119 follows immediately.

a, $\underline{z_f + \frac{h_f}{2} < z \leqslant h}$

Because $L\{p_D\} = \frac{2}{s} L\{Z\}$, we obtain:

$$L\{p_D\} = \frac{4h}{h_f} \frac{\sinh \frac{h_{f_D}}{2}\sqrt{s} \cdot \cosh z_{f_D}\sqrt{s} \cdot \cosh (h - z)_D\sqrt{s}}{s^2 \sinh h_D \sqrt{s}} \tag{120}$$

and this is equivalent to:

$$L\{p_D\} = \frac{h}{h_f} \frac{e^{-\left[z - (z_f + \frac{h_f}{2})\right]_D\sqrt{s}}}{s^2} \tag{121}$$

which vanishes as soon as Eq. 71 (or 72) is satisfied.

The form of Eq. 121 is similar to the one developed in Appendix C for the early time solution for vertical linear flow into the fracture with a storage effect caused by the finite thickness of the fracture, From a comparison of Eq. 121 and Eq. C-16, we can deduce an equivalent storage constant for the fracture:

$$\bar{C}_f = h_{f_D} \tag{122}$$

The form of p_D at early times is obtained by inverting Eq. 121:

$$p_D = \frac{h}{h_f}\left\{\left(t_D + \frac{[z - (z_f + \frac{h_f}{2})]_D^2}{2}\right) \text{erfc} \frac{[z - (z_f + \frac{h_f}{2})]_D}{2 \quad t_D} - [z - (z_f + \frac{h_f}{2})]_D \sqrt{\frac{t_D}{\pi}} \; e^{-\frac{[z - (z_f + \frac{h_f}{2})]_D^2}{4t_D}}\right\} \tag{123}$$

b. $\underline{0 \leqslant z < z_f - \frac{h_f}{2}}$

$$L\{p_D\} = \frac{4h}{h_f} \frac{\sinh \frac{h_{f_D}}{2}\sqrt{s} \cosh (h - z_f)_D\sqrt{s} \cosh z_D\sqrt{s}}{s^2 \sinh h_D\sqrt{s}} \tag{124}$$

which reduces to:

$$L\{p_D\} = \frac{h}{h_f} \frac{e^{-[(z_f - \frac{h_f}{2}) - z]_D\sqrt{s}}}{s^2} \tag{125}$$

when Eq. 77 (or 78) is satisfied. In that case, p_D :is given by:

$$p_D = \frac{h}{h_f}\left\{\left(t_D + \frac{[(z_f - \frac{h_f}{2}) - z]_D^2}{2}\right) \text{erfc} \frac{[(z_f - \frac{h_f}{2}) - z]_D}{2\sqrt{t_D}} - [(z_f - \frac{h_f}{2}) - z]_D \sqrt{\frac{t_D}{\pi}} \; e^{-\frac{[(z_f - \frac{h_f}{2}) - z]_D^2}{4t_D}}\right\} \tag{126}$$

and again, we can deduce the relation:

$$\bar{C}_f = h_{f_D}$$

c. $z_f - \frac{h_f}{2} \leq z \leq z_f + \frac{h_f}{2}$

From Eq. 89, it follows that, inside the fracture:

$$p_D = \frac{2h}{h_f} t_D \tag{127}$$

when the appropriate Eq. 90 is satisfied.

If the pressure is measured at the lower or upper boundary of the fracture, the result may be obtained from Eq. 95:

$$p_D = \frac{h}{h_f} t_D \tag{128}$$

if Eq. 96 is satisfied. This equation is similar to Eq. 127, and shows that the pressure drop on the fracture boundary is one half that within the fracture.

There is an interesting physical interpretation of Eq. 127 giving the pressure within the fracture. Substitution of real variables into Eq. 127 yields:

$$p_i - p = \frac{q_f t}{\pi r_f^2 h_f \phi c} \tag{129}$$

which is a volumetric balance assuming all production originates within the volume of the fracture.

Eq. 129 can also be written as:

$$\frac{2\pi k_r h_f}{q_f \mu} (p_i - p) = 2 \frac{k r t}{\phi \mu c r_f^2} \tag{130}$$

Eq. 130 means that only the fracture is contributing to the flow, because there is no pressure drop above or below the fracture from Eqs. 123 and 126, nor elsewhere in the reservoir because $\lim P\left(\frac{r_D}{\sqrt{2t_D}}, \frac{1}{\sqrt{2t_D}}\right) = 0$ for $r_D > 1$ at the times we consider here.

From Eq. 129, it can also be shown that:

$$\frac{dp}{dt} = -\frac{q_f}{\phi c \pi r_f^2 h_f} \tag{131}$$

which means that at very early times, the rate of decline is constant and proportional to the fluid-filled pore volume of the fracture.

Therefore, at very early times, the flow in fracture is controlled by a storage type flow, which yields a straight line of slope 1 on log-log coordinates. As pointed out in the literature[13], data taken during such a flow cannot be interpreted to yield formation flow capacity. On cartesian coordinates, however, the slope of the straight line corresponding to storage flow is related to the volume of the fracture, as shown by Eq. 131. The duration of the storage flow is limited by the fracture thickness, and by the altitude of the point of measurement in the fracture.

As t_D increases, the term representing the linear vertical flow into the fracture, with a storage constant equal to h_{fD}, becomes important. The duration of the linear flow is limited by the distance between the fracture and the closest reservoir boundary, and by the radial distance to the outer boundary of the fracture,

A transition period follows the linear flow period, during which the pseudo-skin function increases with time until it reaches its maximum value and remains constant, as the flow becomes radial.

11. UNSTEADY-STATE PRESSURE DISTRIBUTION CREATED BY A SINGLE PLANE HORIZONTAL FRACTURE

A. MATHEMATICAL DERIVATION

Horizontal hydraulic fractures are believed to have thicknesses of the order of one tenth of an inch. If the thickness of the reservoir is, say, 50 feet, the ratio $\frac{h_f}{2h}$ is of the order of 10^{-4}, and the sine term in the pseudo-skin function can be assumed equal to its argument,

The sine term can be written as:

$$\sin n\pi \frac{h_f}{2h} = n\pi \frac{h_f}{2h} \left\{ 1 - \frac{n^2\pi^2}{24} \left(\frac{h_f}{h}\right)^2 \right\} + 0 \left\{ \left(\frac{h_f}{h}\right)^5 \right\}$$

We can write:

$$\sin n\pi \frac{h_f}{2h} = n\pi \frac{h_f}{2h}$$

with a maximum error of 1% if:

$$\frac{n^2\pi^2}{24} \left(\frac{h_f}{h}\right)^2 \leqslant \frac{1}{100} \text{ , } n \geqslant 1$$

If we take for convenience:

$$\left(\frac{h_f}{h}\right)^2 \leqslant \frac{25}{100\ \pi^2}$$

The condition is:

$$h_f \leqslant \frac{h}{2\pi} \tag{132}$$

The pressure drop function becomes then:

$$p_D\,(r_D,\ z_D,\ t_D) = \int_0^{t_D} \left[1 + 2 \sum_{n=1}^{\infty} \exp\left(-\frac{n^2\pi^2 t'_D}{h_D^2}\right) \cos n\pi \frac{z_i}{h} \cos n\pi \frac{z}{h}\right]$$

$$\left[\int_0^1 I_o\left(\frac{r_D r'_D}{2t_D}\right) \exp\left(-\frac{r'^2_D}{4t'_D}\right) r'_D\, dr'_D\right] \frac{e^{-\frac{r_D^2}{4t'_D}}}{t'_D}\ dt'_D \tag{133}$$

We recognize in Eq. 133 the pressure drop created in a reservoir by a plane horizontal fracture derived in Appendix B. Therefore, a single horizontal fracture behaves like a plane horizontal. fracture if its thickness is less than one sixth of the formation thickness. This is generally the case for hydraulic fractures,

From the definition of the Z function given in Eq 28, it follows that:

$$Z\left(\frac{t_D}{h_-^2},\ 0,\ \frac{z_f}{h},\ \frac{z}{h}\right) = 1 + 2 \sum_{n=1}^{\infty} \exp\left(-\frac{n^2\pi^2 t'_D}{h_D^2}\right) \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h}$$

and Eq 133 thus becomes:

$$p_D\,(r_D,\, z_D,\, t_D) = 2 \int^{t_D} P\left(\frac{1}{\sqrt{2t'_D}}\quad \frac{r_D}{\sqrt{2t'_D}}\right) Z\left(\frac{t'_D}{h_D^2},\, 0,\, \frac{z_f}{h},\, \frac{z}{h}\right) dt'_D \tag{134}$$

The pseudo-skin function is now defined as:

$$\sigma(r_D,\, z_D,\, t_D) = 4 \sum_{n=1}^{\infty} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h}$$

$$\int_0^{t_D} \exp\left(-\frac{n^2\pi^2 t'_D}{h_D^2}\right) P\left(\frac{1}{\sqrt{2t'_D}},\, \frac{r_D}{\sqrt{2t'_D}}\right) dt'_D \tag{135}$$

or

$$\sigma(r_D,\, z_D,\, t_D) = 2 \int_0^{t_D} P\left(\frac{1}{\sqrt{2t'_D}},\, \frac{r_D}{\sqrt{2t'_D}}\right) \left[Z\left(\frac{t'_D}{h_D^2},\, 0,\, \frac{z_f}{h},\, \frac{z}{h}\right) - 1\right] dt'_D \tag{136}$$

and the pressure drop function given by Eq. 133 can also be written in the form:

$$p_D(r_D,\, z_D,\, t_D) = p_D(r_D,\, t_D) + \sigma(r_D,\, z_D,\, t_D) \tag{137}$$

The properties of a plane horizontal fracture will then be derived from Chapter I by taking the limits of expressions of interest as $h_f \to 0$. All the results obtained in this manner have also been derived directly, but will not be reproduced here as such, because our aim is to show how some new results can be derived from what has been shown in Chapter I for a single horizontal fracture of finite thickness.

Eq. 134 shows that only the Z function has been modified and, therefore, it is only necessary to consider those modifications to study the behavior of the pressure drop function,

B. LONG TIME PRESSURE BEHAVIOR AND WELLBORE PRESSURE

Because the behavior of the Z function at large times is governed by the exponential term,

$$e^{-\frac{n^2\pi^2 t_D}{h_D^2}}$$

it will not be modified if $h_f = 0$.

Therefore, a straight line of slope $\frac{1}{2}$ will be obtained on semi-log coordinates when Eq. 57 and Eq. 98 are satisfied, and the pressure drop function will be given by Eq. 101 and Eq. 102, where the pseudo-skin factor is now given by:

$$\sigma\left(r_D, z_D, \frac{5h_D^2}{\pi^2}\right) = 4 \sum_{n=1} \cos n\pi \frac{}{} \cos n\pi \frac{}{}$$

$$\int_0^{\frac{5}{\pi^2} h_D^2} P\left(\frac{1}{\sqrt{2t_D'}}, \frac{r_D}{\sqrt{2t_D'}}\right) e^{-\frac{n^2\pi^2 t_D'}{h_D^2}} dt_D' \qquad (138)$$

An expression for the pseudo skin factor σ at the wellbore can be derived from Eq. 116 in the following manner.

If we take $z = z_f + \varepsilon x \frac{h_f}{2}$, where $\varepsilon = \pm 1$ and $0 \leqslant x \leqslant 1$, Eq. 116 becomes:

$$\sigma = \frac{2}{3} h_D^2 - \frac{1 + x^2}{4} h_{f_D} h_D - (z_f + \varepsilon x \frac{h_f}{2})_D h_D - \frac{1}{12} h_{f_D}^2$$

$$+ (z_f + \varepsilon x \frac{h_f}{2})_D^2 + z_{f_D}^2$$

$$- \frac{8 h_D^2}{\pi^2 h_{f_D}} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{(z_f + \varepsilon x \frac{h_f}{2})}{h} K_1 (\frac{n\pi}{h_D}) \tag{139}$$

Taking the limit of Eq. 139 as x and $h_f \to 0$, we obtain:

$$\sigma = \frac{2}{3} h_D^2 - 2z_{f_D} (h_D - z_{f_D}) - \frac{4 h_D}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \cos^2 n\pi \frac{z_f}{h} K_1 (\frac{n\pi}{h_D}) \tag{140}$$

For small values of h_D, ($h_D \leqslant 1$), the terms in the infinite sum can be neglected for a precision of 1% and the pseudo skin factor σ is given by:

$$\sigma = \frac{2}{3} h_D^2 - 2z_{f_D} (h_D - z_{f_D}) \tag{141}$$

The wellbore pressure at large times can then be written as:

$$p_{wD} = \frac{1}{2} (Rn\ t_D + 1.80907 + 20) \qquad (142)$$

When the fracture is at the center, top or bottom of the formation, the pseudo-skin factor becomes:

$$\sigma = h_D \left[\frac{1}{3} \left(\frac{h_D}{2}\right) - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} K_1 \left(\frac{n\pi}{h_D/2}\right) \right] , \text{ if } z_f = \frac{h}{2} \qquad (143)$$

and

$$\sigma = 2h_D \left[\frac{1}{3} h_D - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} K_1 \left(\frac{n\pi}{h_D}\right) \right] , \text{ if } z_f = 0 \text{ or } z_f = h \qquad (144)$$

A plot of σ from **Eq.** 143 (center of formation) is shown in Fig. 5.

Only one other expression appears in the literature for the transient pressure drop created by a plane horizontal fracture of infinite conductivity. **It** has been derived by J. E. Warren[15] from a study of bottom-water-drfve reservoirs by K. E. Coats[16] and represents an average value over the area of the fracture. If we define the average pressure in the fracture by:

$$p_{av} = \frac{\int_0^1 p_D(r_D, z_D, t_D)\ 2\pi\ r_D\ dr_D}{\pi}$$

we obtain at large times, from **Eq.** 101:

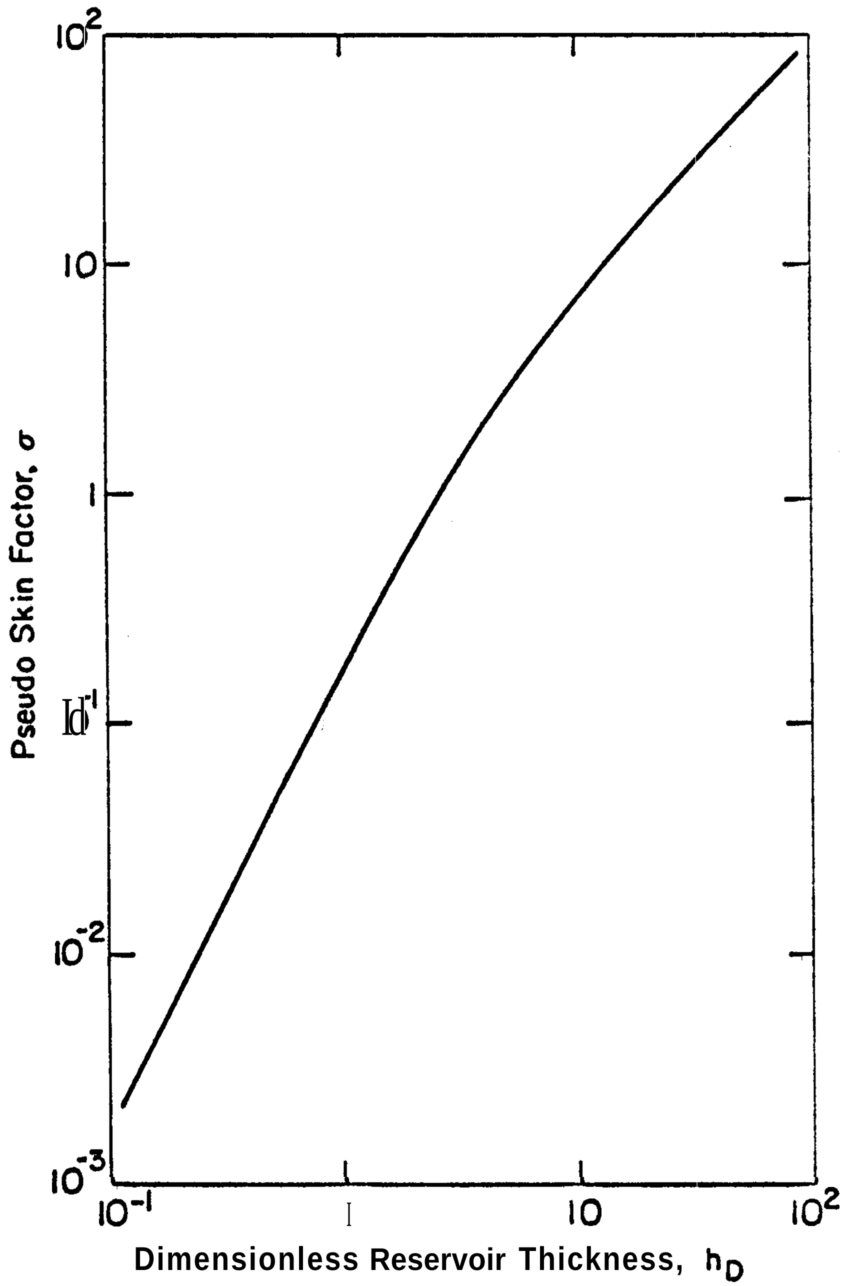

Fig. 5. Pseudo skin factor σ at wellbore (Plane horizontal fracture at center of formation)

$$p_{av} = \frac{1}{\pi} \int_0^1 \frac{1}{2} (\ell n\ t_D + 1.80907 - r_D^2)\ 2\pi\ r_D\ dr_D + \sigma_{av} \tag{145}$$

where the average pseudo skin factor is given by:

$$\sigma_{av} = \frac{\int_0^1 \sigma\left(r_D,\ z_{f_D},\ \frac{5h_D^2}{\pi^2}\right) 2\pi\ r_D\ dr_D}{\pi} \tag{146}$$

Substitution of **Eq.** 136 fnto **Eq.** 146 yields:

$$\sigma_{av} = 2 \int_0^{\frac{5h_D^2}{\pi^2}} \left\{ \frac{\int_0^1 P\left(\frac{1}{\sqrt{2t_D'}},\ \frac{r_D}{\sqrt{2t_D'}}\right)\ 2\pi\ r_D\ dr_D}{\pi} \right\} \left[Z\left(\frac{t_D}{h_D^2},\ 0,\ \frac{z_f}{h},\ \frac{z_f}{h}\right) - 1 \right] dt_D' \tag{147}$$

$$\frac{\int_0^1 P\left(\frac{1}{\sqrt{2t_D}},\ \frac{t_D}{\sqrt{2t_D}}\right) 2\pi\ r_D\ dr_D}{\pi}$$

represents the fractional coverage C_P by a circular area of radius **1** centered at $r_D = 0$ of an infinite plane, weighted radially as a P function, Using a notation due to Germond41 we write:

$$C_P = \frac{S\left(\frac{1}{\sqrt{2t_D}},\ \frac{1}{\sqrt{2t_D}}\right)}{\pi\left(\frac{1}{\sqrt{2t_D}}\right)^2} \tag{148}$$

C_p has been tabulated by Germond[41] and is also plotted as a funcrion of its parameters in Appendix II of Reference 26, Eq 145 then becomes:

$$p_{av} = \frac{1}{2} (\ln t_D + 0,80907) + \frac{1}{4} + \sigma_{av} \tag{149}$$

We can write Warren's results[15] in a similar way as:

$$(p_{av})_W = \frac{1}{2} (\ln t_D + 0.80907) + \frac{1}{4} + (\sigma_{av})_W \tag{150}$$

where

$$(\sigma_{av})_W = \begin{cases} h_D \, A(\frac{h_D}{2}) - 0.6545, \text{ if } z_f = \frac{h}{2} & (151) \\ 2h_D \, A(h_D) - 0.6545, \text{ if } z_f = 0 \text{ or } z_f = h & (152) \end{cases}$$

in which $A(h_D)$ is a geometrical constant given in Table 2 of Coats' paper, A comparison has been made in Fig. 6 between the numerical values of the average pseudo-skin factors from this study and Ref. 15, for different values of rhe dimensionless reservoir thickness h_D. There is excellent agreement"

For the steady state case, values of the skin factor at the wellbore have been published by J. H. Hartsock and J. E. Warren[14], from which it is possible to deduce a pseudo-skin factor $\sigma_{H,W}$, by adding the quantity $\ln r_f/r_w$. Hartsock and Warren's expression for %hewellbore pressure for a fracture at the center of the formation can be written as:

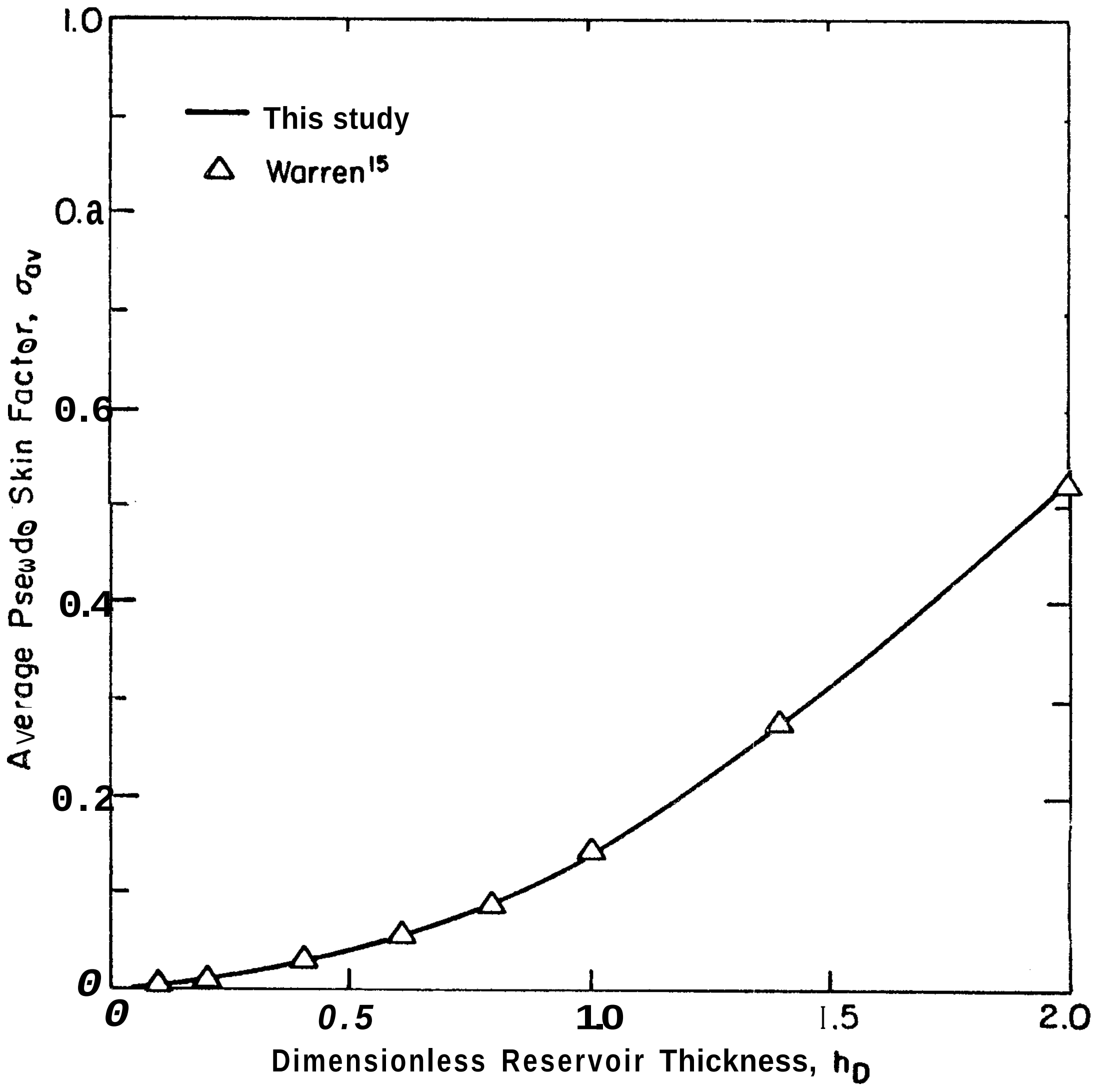

Fig. 6. Average pseudo-skin factor σ_{av} vs. h_D for a single plane horizontal fracture at center of formation

$$p_{wD} = \frac{1}{2}\left(\ln t_D + 1.80907 + 2\sigma_{H,W}\right) \tag{153}$$

where:

$$\sigma_{H,W} = s_{H,W} + \ln \frac{r_f}{r_w} - \frac{1}{2}\,, \tag{154}$$

$s_{H,W}$ being Hartsock and Warren's calculated skin effect.

A comparison has been made between the numerical values taken at the wellbore by the pseudo skin factor σ from Eqs. 143 and 154, for different values of the dimensionless reservoir thickness h_D, when the fracture is at the center of the formation. The comparison is shown in Fig. 7. The important differences between the two results can be explained by the approximations made in Hartsock and Warren's numerical study.

The radius of influence is the same for a plane fracture as for a fracture of finite thickness and is given by Eq. 117 or 118. The existence of a radius of influence has also been considered by Hartsock and Warren, but assumed equal to four times the fracture radius, which thus assumes implicitely that:

$$h_D = 0.3\pi$$

Because the value of the radius of influence is determined by the precision we require for the numerical computations, we can conclude that correct results are obtained only when h_D is between unity and two in Hartsock and Warren's case. This seems to agree with the comparison shown in Fig. 7.

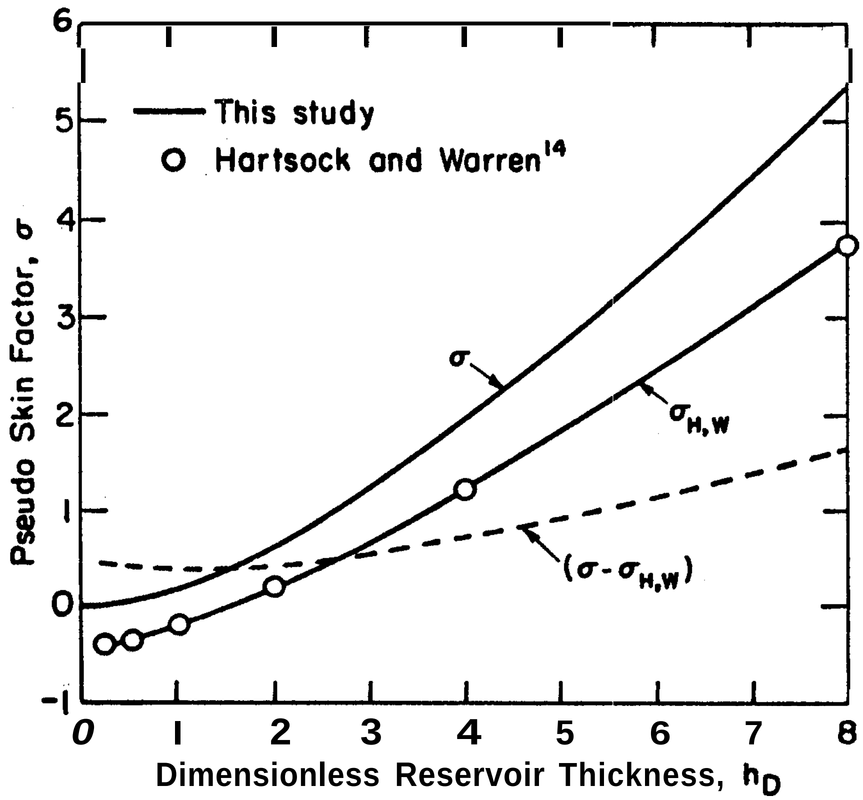

Fig. 7. Wellbore pseudo-skin factor σ vs. h_D for a single plane horizontal fracture at center of formation

C. SHORT TIME PRESSURE BEHAVIOR

The short time behavior of the pressure drop function is obtained from Chapter I, section G, by taking the limit of expressions of interest as $h_f \to 0$.

a. $z_f < z \leqslant h$

From Eq. 120:

$$L\{p_D\} = 2h_D \frac{\cosh z_{f_D}\sqrt{s} \cdot \cosh (h - z)_D\sqrt{s}}{s\sqrt{s}\sinh h_D\sqrt{s}} \tag{155}$$

which is equivalent to:

$$L\{p_D\} = h_D \frac{e^{-(z - z_f)_D\sqrt{s}}}{s^{3/2}} \quad \text{as } s \to \infty \tag{156}$$

Inversion leads to:

$$p_D = 2h_D\sqrt{\frac{t_D}{\pi}}\; e^{-\frac{(z - z_f)_D^2}{4t_D}} - h_D\,(z - z_f)_D \operatorname{erfc}\frac{(z_f - z)_D}{2\sqrt{t_D}} \tag{157}$$

Eq. 157 represents a linear vertical flow into the fracture, and p_D is negligible when:

$$t_D \leqslant \frac{(z - z_f)_D^2}{20} \tag{158}$$

b. $0 \leqslant z < z_f$

Similarly, from Eq. 124:

$$L\{p_D\} = 2h_D \frac{\cosh (h - z_f)_D\sqrt{s} \cos z_D\sqrt{s}}{s\sqrt{s} \sinh h_D\sqrt{s}} \tag{159}$$

At early times:

$$L\{p_D\} = h_D \frac{e^{-(z_f - z)_D\sqrt{s}}}{s^{3/2}} \tag{160}$$

After inversion:

$$p_D = 2h_D\sqrt{\frac{t_D}{\pi}}\ e^{-\frac{(z_f - z)^2}{4t_D}} - h_D\ (z_f - z)_D \operatorname{erfc} \frac{(z_f - z)_D}{2\sqrt{t_D}} \tag{161}$$

which represents linear vertical flow into the fracture and is negligible when:

$$t_D \leqslant \frac{(z_f - z)_D^2}{20} \tag{162}$$

c. $\underline{z = z_f}$

When Eq. 119, in which h_f is set to zero, represents the pressure drop function created by the plane fracture,, the pressure drop at the fracture is given by:

$$p_D = 2h_D\sqrt{\frac{t_D}{\pi}} \tag{163}$$

Therefore, at very early times, the flow into the fracture is vertical and linear. As time increases, the duration of this flow will be limited by the existence of the upper or lower boundary and by the fracture radius, which determines the time when the P function is no longer constant. Numerical results show that the upper and lower boundary effects are not felt at the wellbore when $h_D \geqslant 1$, and therefore, the duration of the linear flow is only limited by the fracture radius. When $h_D < 1$, the duration of the linear flow is only limited by the upper and lower boundary effects, whose importance is then overcome by the variation of the P function, indicating the beginning of the radial flow. A transition period follows, until the radial flow is reached as the pseudo skin function becomes constant.

III. UNSTEADY-STATE PRESSURE DISTRIBUTION CREATED BY A WELL WITH RESTRICTED FLOW ENTRY OR PARTIAL PENETRATION

The effects of partial penetration and restricted fluid entry have been studied analytically[21, 27, 42-46] or numerically[47] by a number of authors, and solutions have been given for early-transient, transient and the steady state[42, 44] flow.

The subject is again treated here to show how the solution can be derived from Eq. 26 representing the pressure drop created by a single horizontal fracture of thickness h_f. Many features cf partial penetration and restricted flow entry follow immediately, when we express all dimensionless quantities with respect to r_w instead of r_f.

A. MATHEMATICAL DERIVATION

When $r_f \to 0$, the solid cylinder source becomes a line source, and a single fracture becomes a limited entry, restricted to the thickness h_f. In these cases, we only consider the points in the reservoir for which $r_D \geqslant 1$, and the pressure at the well is obtained by using $r_D = 1$.

From Eq. A-9 in Appendix A, it can be shown that, as $r_f \to 0$, the P function defined by Eq. 27, can be approximated by:

$$P\left(\frac{1}{\sqrt{2t_D}}, \frac{r_D}{\sqrt{2t_D}}\right) = \frac{e^{-\frac{r_D^2}{4t_D}}}{4t_D} \tag{164}$$

and the solution for a well partially penetrating, or with restricted flow entry is obtained from Eq. 26 as:

$$p_D\ (r_D,\ z_D,\ t_D) = \int^{t_D} Z\left(\frac{t'_D}{h_D^2}\ ,\ \frac{h_f}{2h}\ ,\ \frac{z_f}{h}\ ,\ \frac{z}{h}\right)\frac{e^{-\frac{r_D^2}{4t'_D}}}{\quad}\ dt'_D \tag{165}$$

Setting $u = \dfrac{r_D^2}{4t_D}$, and using the exponential integral notation, Eq. 165 becomes:

$$p_D\ (r_D,\ z_D,\ tD) = -\frac{1}{2}\,Ei\left(-\frac{r_D^2}{4t_D}\right)$$

$$+\ \frac{4h}{\pi h_f}\sum_{n=1}^{\infty}\ \frac{1}{n}\ \sin\ n\pi\ \frac{h_f}{2h}\ \cos\ n\pi\ \frac{z_f}{h}\ \cos\ n\pi\ \frac{z}{h}$$

$$\int_{\frac{r_D^2}{4t_D}}^{\infty} e^{-\left[u\ +\ \frac{\left(\frac{n\pi\ r_D}{h_D}\right)^2}{4u}\right]}\ \frac{du}{u} \tag{166}$$

Eq. 166 is identical to the solution derived by Hantush[27] by using Laplace and Fourier transforms.

As we have done before, we write Eq. 165 or 166 in the form:

$$p_D(r_D, z_D, t_D) = p_D(r_D, t_D) + s(r_D, z_D, t_D), \qquad r_D \geqslant 1 \tag{167}$$

where

$$p_D(r_D, t_D) = -\frac{1}{2}\,\mathrm{Ei}\left(-\frac{r_D^2}{4t_D}\right)$$

represents the line source solution, and:

$$s(r_D, z_D, t_D) = \frac{4h}{\pi h_f}\sum_{n=1}^{\infty}\frac{1}{n}\sin n\pi\frac{h_f}{2h}\cos n\pi\frac{z_f}{h}\cos n\pi\frac{z}{h}$$

$$\int_{\frac{r_D^2}{4t_D}}^{\infty} e^{-\left[u + \frac{\left(\frac{n\pi\, r_D}{h_D}\right)^2}{4u}\right]}\frac{du}{u}$$

represents the skin function and is related to the skin factor S defined at the wellbore ($r_D = 1$) by Van Everdingen[32] and Hurst[33].

B. LONG TIME PRESSURE BEHAVIOR

The long time behavior of the pressure drop function given by Eq. 167 can be derived directly as in Chapter I, or deduced from the results of Chapter I as was done in Chapter II.

Plotting the pressure drop function versus the natural logarithm of time gives a curve whose slope is:

$$\frac{dp_D\ (r_D,\ z_D,\ t_D)}{d(\ell n\ t_D)} = \frac{1}{2}\ Z\left(\frac{t_D}{h_D^2}\ ,\ \frac{h_f}{2h}\ ,\ \frac{z_f}{h}\ ,\ \frac{z}{h}\right)\ \exp\left(-\ \frac{r_D^2}{4t_D}\right) \tag{168}$$

A straight line of slope $\frac{1}{2}$ is obtained when Eq. 98 is satisfied, and as soon as $\exp\left(-\ \frac{r_D^2}{4t_D}\right)$ can be taken as unity. That is, when:

$$t_D \geqslant 25\ r_D^2 \tag{169}$$

Eq. 169 could have been deduced from Eq. 57 by writing it as:

$$\frac{k_r\ t}{\phi\mu c} \geqslant 12.5\ (2r^2 + r_f^2)$$

Because

$$\lim_{r_f \to 0}\ (2r^2 + r_f^2) = 2r^2$$

we are left with:

$$t_D \geqslant 25\ r_D^2$$

Eq. 166 then becomes:

$$p_D\ (r_D,\ z_D,\ t_D) = \frac{1}{2}\ [\ell n\ \frac{t_D}{r_D^2} + 0.80907 + 2s(r_D,\ z_D,\ \frac{5h_D^2}{\pi^2})] \tag{170}$$

where $s(r_D, z_D, \frac{5h_D^2}{\pi^2})$ is the maximum value of the skin. function $s(r_D, z_D, t_D)$ and can be shown [27, 40] to be equal to:

$$s(r_D, z_D, hD) = \frac{4h}{\pi h_f} \sum_{n=1}^{\infty} \frac{1}{n} \sin n\pi \frac{h_f}{2h} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h} K_o\left(\frac{n\pi\ r_D}{h_D}\right) \tag{171}$$

The value of the radius of influence follows immediately because $s(r_D, z_D, \frac{5h_D^2}{\pi^2})$ is negligible when $K_o\ (\frac{n\pi\ r_D}{h_D})$ is negligible. That is, when:

$$r_D \geqslant \frac{10}{\pi} h_D$$

as can be derived from **Eq.** 118.

C. SHORT TIME PRESSURE BEHAVIOR

The short time behavior follows from the study of the Z function:

a. $z_f + \frac{h_f}{2} < z \leqslant h\ \ (\text{or } 0 \leqslant z < z_f + \frac{h_f}{2})$

From **Eq.** 70 (or **Eq.** 76), the **Z** function is negligible when **Eq.** 71 (or Eq. 77) is satisfied, as is the pressure drop function.

b. $z_f - \frac{h_f}{2} \leqslant z \leqslant z_f + \frac{h_f}{2}$

From **Eq.** 89, the pressure drop function can be approximated by

$$p_D = \int_0^{t_D} \frac{h}{h_f} \frac{e^{-\frac{r_D^2}{4t_D'}}}{2t_D'}\ dt_D'\ ,$$

or:

$$p_D = -\frac{1}{2}\frac{h}{h_f}\,Ei\left(-\frac{r_D^2}{4t_D}\right), \tag{172}$$

when **Eq.** 91 is satisfied, if $(z_f - \frac{h_f}{2}) < z < (z_f + \frac{h_f}{2})$, and by:

$$p_D = -\frac{1}{4}\frac{h}{h_f}\,Ei\left(-\frac{r_D^2}{4t_D}\right) \tag{173}$$

when Eq. 96 is satisfied, if $z = z_f - \frac{h_f}{2}$ or $z = z_f + \frac{h_f}{2}$.

Eq. 172 can be written as:

$$\frac{2\pi\, k_r\, h_f}{q_f\, \mu}(p_i - p) = -\frac{1}{2}\,Ei\left(-\frac{r_D^2}{4t_D}\right), \tag{174}$$

which shows that the well behaves as if the total sand thickness were equal to h_f.[43]

When t_D is larger than the limits given by Eqs. 71, 77, 91 or 96, there exists a transition period during which the skin function varies with time, until it becomes constant when the transient period begins. It has then reached its maximum value and is given by **Eq.** 171.

IV. UNSTEADY STATE PRESSURE DISTRIBUTION CREATED BY A WELL AND A SINGLE HORIZONTAL FRACTURE OF FINITE CONDUCTIVITY AT CENTER OF FORMATION. FINITE ELEMENT METHOD

The problem of a reservoir producing at constant rate through a well and a horizontal fracture of finite conductivity at the center of the formation has been investigated numerically with the finite element method, This method was first applied to petroleum engineering problems involving single well flow problems by Javandel and Witherspoon[48]. A computer program provided by S. P. Neumanand P. A, Witherspoon of the University of California was slightly modified by the writer and used in rhe present problem.

With this problem, it was found that the finite element method was extremely sensitive to the array and the number of mesh points used in the computation. No reliable results were obtained for dimensionless times , t_{DW}, less than 2,000. Several runs were made with increasing numbers of mesh points, from 242 to 316. The mesh, as shown in Fig. 8, was designed to be small where the pressure gradients were high, and large where the pressure gradients were small. Yet none of the interesting features shown by the analytical solution at early times could be located. This seems to be partly a result of the specffic problem which necessitated very small mesh spacings, and partly a result of the method itself, which involves a solution of linear equations which is not accurate at early times (storage and linear flow periods).

The results obtained after dimensionless times, t_{DW}, greater than 2,000 have been plotted on a semi-log graph and appear in Fig. 9.

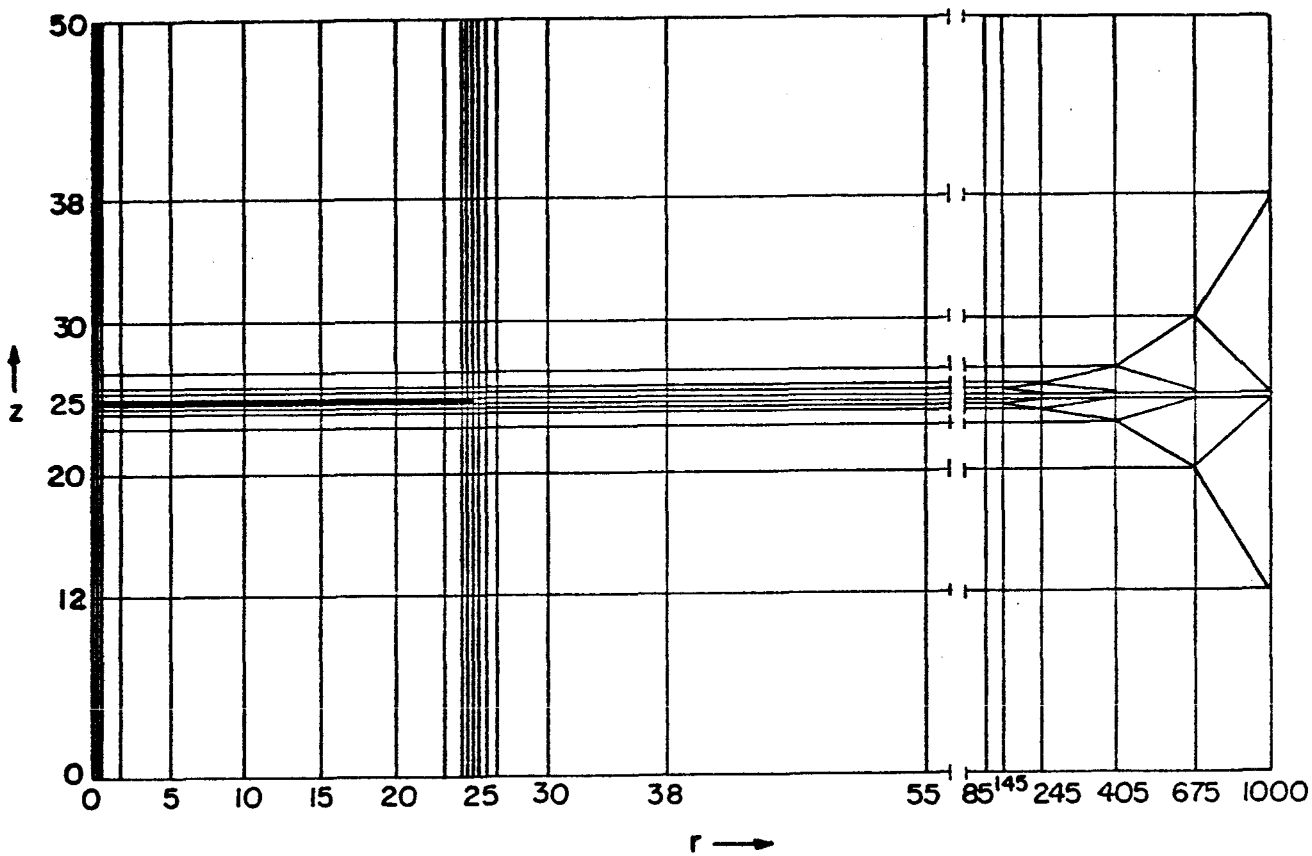

Fig. 8. Network system for axisymmetric flow to well of zero radius and single horizontal fracture at center of formation

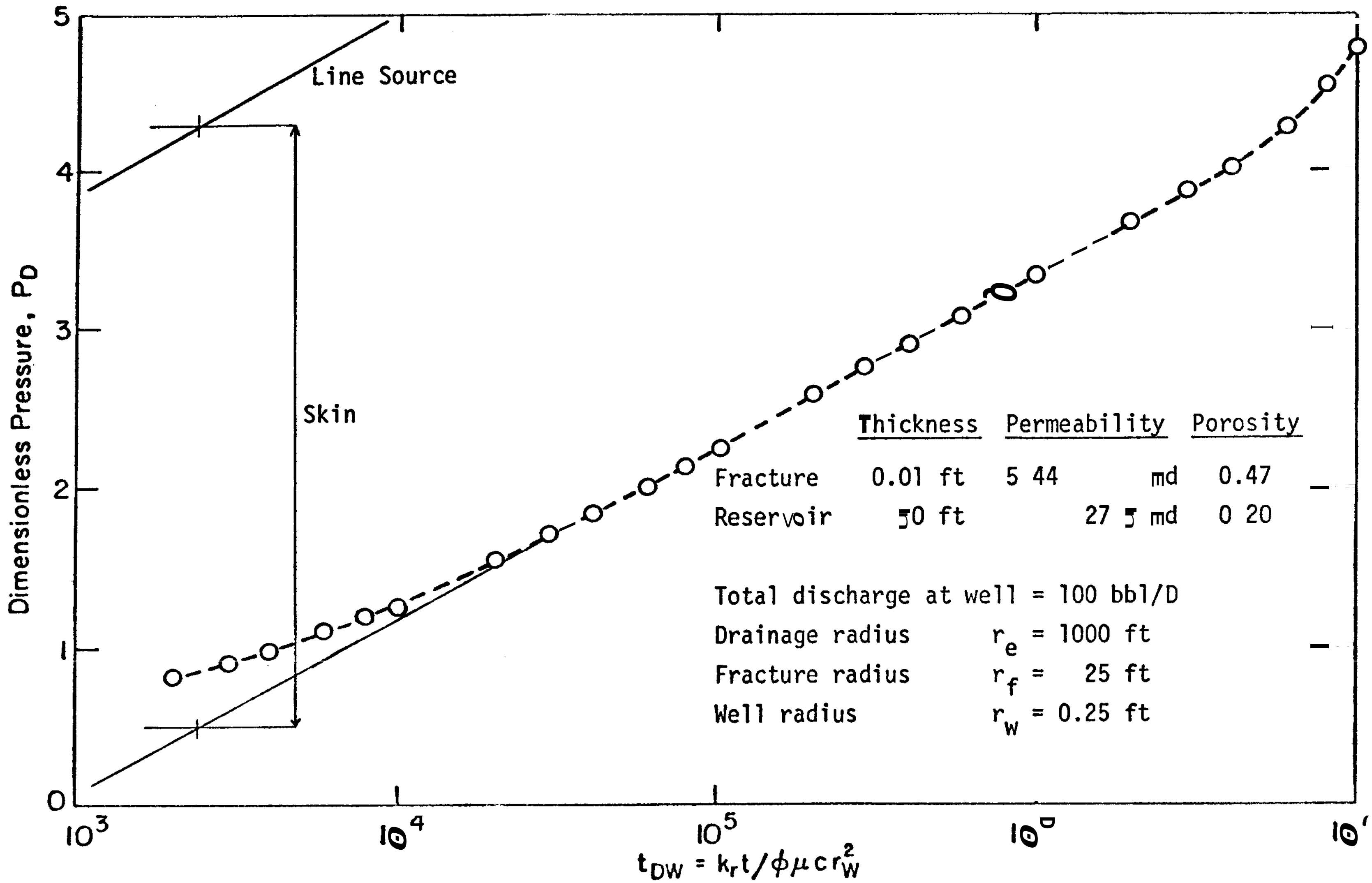

Fig. 9. Finite element method: fracture at center of formation

CONCLUSIONS

An analytical study of the pressure distribution created by a single horizontal fracture in an infinite reservoir with impermeable upper and lower boundaries has shown that there exists four different flow periods.

A storage type flow occurs first, the duration of which is limited by the fracture thickness, and by the shortest vertical distance from the point at which the pressure is measured to the fracture boundaries.

A vertical linear type flow from the reservoir into the fracture follows, and the duration of this period is limited by the shortest vertical distance from the fracture to the reservoir boundaries, and by the radial distance from the point to the fracture outer boundary.

After a period of transition, the transient flow period starts at a time depending upon the radial distance from the point of measurement to the wellbore axis, and upon the reservoir thickness. The flow is radial and the pressure is the same as that created by a line source well with a constant skin. The skin effect is independent of time but does depend upon the position of the pressure point and is negligible at distances that are greater than three times the reservoir thickness.

If the fracture thickness is less than one sixth of the formation thickness, the fracture behaves like a single plane horizontal fracture; the storage type flow no longer exists and only the last three flow periods are left, with the same time limits.

The same mathematical model can be applied to derive the pressure distribution created by a well with restricted flow entry or created by a well that only partially penetrates the formation. Again, three different flow periods can be characterized: (1) an early radial flow into the open zone, the duration of which is limited by the height of rhe opening, and by the altitude of the point of measurement in the open interval; (2) a transition period, and (3) a transient radial flow period as in the fracture case. These results are summarized in Fig. PO,

The time at which the radial flow period starts is the same in all cases, but rhe skin is different in each case. Analysis of the pressure build-up data san thus be made in a conventional manner if the scraighr line has been obtained[39], or as recommended by H. J. Ramey[13] if early time data are available,

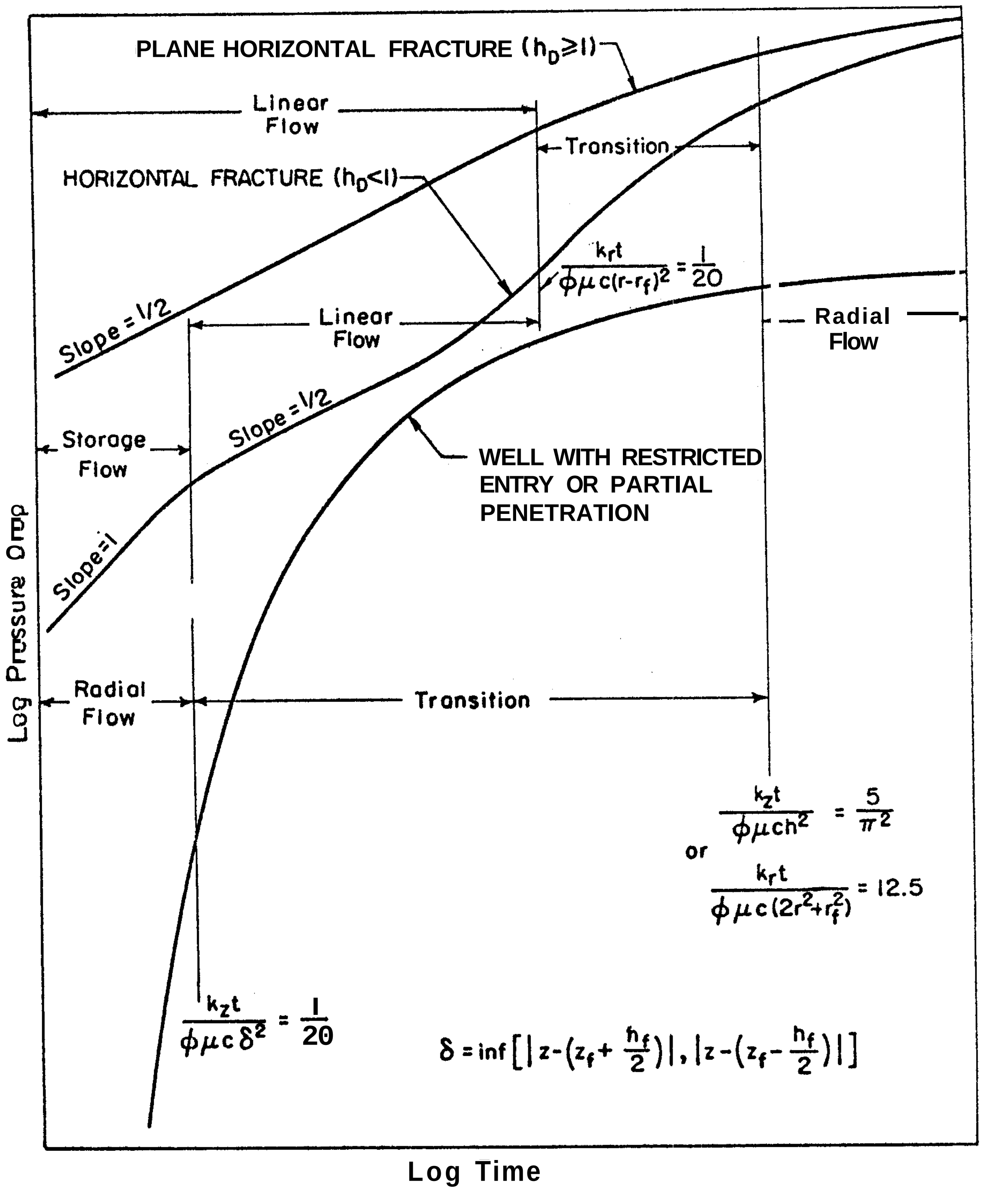

Fig. 10. **Illustration of different flow periods** for horizontal fractures **limited flow entry** and **partial penetratiion**

NOMENCLATURE

The Darcy system of units[18] has been used.

c = compressibility

C = storage constant

$\bar{C}$ = dimensionless storage constant

C_P = fractional coverage by the P function of a circular area

d = distance between two points

h = formation thickness

h_f = fracture thickness

k = formation permeability

p_i = initial reservoir pressure

p_{wD} = dimensionless wellbore pressure drop

p_D = dimensionless pressure drop

Δp = pressure drop

q = instantaneous point source strength; production rate of well

q_f = production rate of fracture

r_D = dimensionless radius

r_{wD} = dimensionless wellbore radius

r_w = wellbore radius

s = skin factor; variable in Laplace transformation

t = time of flowing

t_D = dimensionless time of flowing based on r_f

z = vertical distance to the reservoir lower boundary

γ = Euler's constant = 0.5772

η = diffusivity constant

μ = viscosity

ϕ = porosity

σ = pseudo-skin factor

Subscripts

av = areal average

D = dimensionless

f = fracture

i = initial; influence

w = well

Special functions

$$\text{erf}\,(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-u^2}\,du$$ Error function

$$\text{erfc}\,(x) = 1 - \text{erf}\,(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-u^2}\,du$$ Complementary error function

$$-\text{Ei}\,(-x) = \int_x^\infty \frac{e^{-u}}{u}\,du$$ Exponential integral

$I_o(x)$ Modified Bessel function of the first kind of order 0

$I_1(x)$ Modified Bessel function of the first kind of order 1

$K_o(x)$ Modified Bessel function of the second kind of order 0

$K_1(x)$ Modified Bessel function of the second kind of order 1

REFERENCES

1. Matthews, C. S. and Russell, D. G.: Pressure Build-up and Flow Tests in Wells, Soc. Pet. Eng. Monograph Series, Vol. 1, SPE, Dallas (1967) 103.

2. Zemanek, J., Caldwell, R. L., Glenn, E. E., Jr., Holcomb, S. V., Norton, L. J. and Strauss, A. J. D.: "The Borehole Televiewer -- A New Logging Concept for Fracture Location and Other Types of Borehole Inspection", J. Pet. Tech. (June, 1969) 762-774.

3. Prats, M. "Effect of Vertical Fractures on Reservoir Behavior -- Incompressible Fluid Case", Soc. Pet. Eng. J. (June, 1961) 105-118.

4. Prats, M., Hazebroek, P. and Strickler, W. R.: "Effect of Vertical Fractures on Reservoir Behavior -- Compressible-Fluid Case", Soc. Pet. Eng. J. (June, 1962) 87-94,

5. Van Everdingen, A. F. and Joffre Meyer, L.: "Analysis of Buildup Curves Obtained after Well Treatment", Preprint of paper SPE 2864 presented at 9th Biennal Production Techniques Symposium (May 14-15, 1970) in Wichita Falls,

6. McGuire, W. J. and Sikora, V. J.: "The Effect of Vertical Fractures on Well Productivity", Trans., AIME (1960) 219, 401-403.

7. Dyes, A. B., Kemp C. E. and Candle, B. H.: "Effect of Fractures on Sweep-Out Pattern", Trans., AIME (1958) 213, 245-249.

8. Scott, J. O.: "The Effect of Vertical Fractures on Transient Pressure Behavior of Wells", J. Pet. Tech, (Dec., 1963) 1365-1369.

9. Russell, D. G. and Truitt, N. E.: "Transient Pressure Behavior in Vertically Fractured Reservoirs", J. Pet. Tech. (Oct., 1964) 1159-1170.

10. Wattenbarger, R. A, and Ramey, H. J., Jr.: "Well Test Interpretation of Vertically Fractured Gas Wells", J. Pet. Tech. (May, 1969) 625-632.

11. Millheim, Keith K. and Cichowicz, Leo: "Testing and Analyzing Low-Permeability Fractured Gas-Wells", J. Pet. Tech. (Feb., 1968) 193-198.

12. Russell, D. G.: "Extensions of Pressure Build-up Analysis Methods", J. Pet. Tech. (Dec., 1966) 1624-1636.

13. Ramey, H. J., Jr.: "Short-Time Well Test Data Interpretation in the Presence of Skin Effect and Wellbore Storage", J. Pet. Tech. (Jan., 1970) 97-104.

14. Hartsock, J. H. and Warren, J. E.: "The Effect of Horizontal Hydraulic Fracturing on Well Performance", J. Pet. Tech. (Oct., 1961) 1050-1056.

15. Warren, J. E.: "Discussion on a Mathematical Model for Water Movement about Bottom-Water-Drive Reservoirs", Soc. Pet. Eng. J. (Dec., 1962) 367-368.

16. Coats, K. H.: "A Mathematical Model for Water Movement about Bottom-Water-Drive Reservoirs", Soc. Pet. Eng. J. (March, 1962) 44-52.

17. Hubbert, M. K.: "Darcy's Law and the Field Equations of the Flow of Underground Fluids", Trans., AIME (1956) 207, 222-239.

18. Matthews, L. S. and Russell, D. G.: op. cit., 4-17.

19. Lord Kelvin (Sir William Thomson): Mathematical and Physical Papers, Cambridge at the University Press (1884) II, 41.

20. Carslaw, H. S. and Jaeger, J. C.: Conduction of Heat in Solids, Oxford at the Clarendon Press (1959) 255-256.

21. Nisle, R. G.: "The Effect of Partial Penetration on Pressure Build-up in Oil Wells", Trans., AIME (1958) 213, 85-90.

22. Carslaw, H. S. and Jaeger, J. C.: op. cit., 482-484.

23. Carslaw, H. S. and Jaeger, J. C.: op. cit., 273.

24. Carslaw, H. S. and Jaeger, J. C.: op. cit., 275-276.

25. Carslaw, H. S. and Jaeger, J. C.: op. cit., 260.

26. Masters, J. I.: "Some Applications in Physics of the P Function", J. Chem. Phys. 23 (1955) 1865-74.

27. Hantush, M. S. "Nonsteady Flow to a Well Partially Penetrating an Infinite Leaky Aquifer", Proceedings, Iraq Scientific Society, 1957.

28. Miller, F. G.: "Theory of Unsteady-State Influx of Water in Linear Reservoirs", J. Pet. Institute (Nov., 1962) 48, 365-379.

29. Carslaw, H. S. and Jaeger, J. C.: op. cit., 353.

30. Hantush, M. S. and Jacob, C. E.: "Non-Steady Green's Function for an Infinite Strip of Leaky Aquifer", Trans., Amer. Geophys. Union, 36, (1955) 101.

31. Ramey, H. J., Jr." "Application of the Line Source Solution to Flow in Porous Media - A Review", Prod. Monthly, 31, No. 5 (1967) 4,

32. Van Everdingen, A. F.: "The Skin Effect and its Influence on the Productive Capacity of a Well", Trans., AIME (1953) 198, 171-176.

33. Hurst, W.: "Establishment of the Skin Effect and its Impediment to Fluid Flow into a Wellbore", Pet. Eng. (Oct., 1953) 25, B-6.

34. Carslaw, H. S. and Jaeger, J. C.: op. cit., 494-496.

35. Carslaw, H. S., and Jaeger, J. C.: op. cit., 488-490.

36. Abramowitz, M. and Stegun, I. A.: Handbook of Mathematical Functions, U. S. Dept. of Commerce, A. M. S. 55 (1967) 374-380.

37. Gradshteyn, I. S. and Ryzhik, I. M.: Tables of Integrals, Series and Products, Academic Press, New York, London, 4th Ed. (1965) 717.

38. Gradshteyn, I. S. and Ryzhik, I. M.: op. cit., 38-41.

39. Matthews, C. S. and Russel, D. G.: op. cit., 4-17.

40. Gradshteyn, I. S. and Ryzhik, I. M.: op. cit., 340 No. 9.

41. Germond, H. H.: "The Circular Coverage Function", R-M-330, January 26, 1950 (copies of this report may be obtained from the Rand Corporation, Santa Monica, California).

42. Muskat, M.: The Flow of Homogeneous Fluids Through Porous Media, J. W. Edwards, Inc., Ann Arbor, Michigan (1946) 243.

43. Brons, F. and Marting, V. E.: "The Effect of Restricted Fluid Entry on Well Productivity", J. Pet. Tech, (Feb., 1961) 172.

44. Odeh, A. S.: "Steady-State Flow Capacity of Wells with Limited Entry to Flow", Soc. Pet. Eng. J. (March, 1968) 43-51.

45. Seth, M. S.: "Unsteady-State Pressure Distribution in a Finite Reservoir with Partial Well Bore Opening", J. Canadian Pet, Tech. (Oct.-Dec., 1968) 7, 153.

46. Clegg, M. W. and Mills, M.: "A Study of the Behavior of Partially Penetrating Wells", Soc. Pet. Eng. J. (June, 1969) 189.

47. Kazemi, H. and Seth, M. S.: "Effect of Anisotropy and Stratification on Pressure Transient Analysis of Wells with Restricted Flow Entry", J. Pet. Tech. (May, 1969) 639-647.

48. Javandel, I. and Witherspoon, P. A.: "Application of the Finite Element Method to Transient Flow in Porous Media", Soc. Pet. Eng. J. (Sept., 1968) 241.

49. Carslaw, H. S. and Jaeger, J. C.: op. cit., 316.

50. Ralston, A.: A First Course in Numerical Analysis, McGraw-Hill Book Company, Inc., New York (1965) 121.

51. Van Everdingen, A. F. and Hurst, W.: "The Application of the Laplace Transformation to Flow Problems in Reservoirs", Trans., AIME (1949) 186, 305-324.

52. Edwardson, M. J., Girner, H. M., Parkinson, H. R., Williams, C. D. and Matthews, C. S.: "Calculation of Formation Temperature Disturbances Caused by Mud Circulation", J. Pet. Tech. (April, 1962) 416.

APPENDIX A

Unsteadv-State Pressure Distribution Created by a Solid Cylinder Source

We use the instantaneous point source solution defined by Eq. 6 in the text to generate a cylinder source of infinite height: the pressure distribution created by such a cylinder of radius r is obtained by integrating Eq. 7 with respect to z from $-\infty$ to $+\infty$. The result is:

$$\Delta p^{(1)} = \frac{q\ r\ dr\ dt}{4\sqrt{\pi\phi c}\ (\eta t)^{3/2}} \exp\left(-\frac{r_m^2 + r^2}{4\eta t}\right) I_o\left(\frac{r_m\ r}{2\eta t}\right)$$

$$\int_{-\infty}^{+\infty} e^{-\frac{(z - z_m)^2}{4\eta t}}\ dz$$

which reduces to:

$$\Delta p^{(1)} = \frac{q\ r\ dr\ dt}{2\phi c\eta t}\ I_o\left(\frac{r_m\ r}{2\eta t}\right) \exp\left(-\frac{r_m^2 + r^2}{4\eta t}\right) \qquad \text{(A-1)}$$

The instantaneous solid cylinder of radius r_f is then obtained by integrating Eq. A-1 with respect to r from 0 to r_f.

$$\Delta p = \frac{q\ dt}{2\phi c\ \eta t}\ e^{-\frac{r_m^2}{4\eta t}} \int_0^{r_f} I_o\left(\frac{r_m\ r}{2\eta t}\right) e^{-\frac{r^2}{4\eta t}}\ r\ dr \qquad \text{(A-2)}$$

q is the constant withdrawal rate per unit of time and per unit of volume.

The pressure drop creared by a continuous solid cylinder source is then given by:

$$\Delta p = \frac{q}{2\phi c\ \eta} \int_0^t \left[\int_0^{r_f} I_o \left(\frac{r_m\ r}{2\eta\ (t - t')} \right) e^{-\frac{r^2}{4\eta\ (t - t')}}\ r\ dr \right]$$

$$e^{-\frac{r_m^2}{4\eta\ (t - t')}}\ \frac{dt'}{t - t'} \qquad \text{(A-3)}$$

Using t' = t - t' in the integrand of Eq. *A-3,* and noting that:

$$q_w = \pi\ r_f^2\ h\ q\ , \qquad \text{(A-4)}$$

The final form of the solid cylinder pressure drop is:

$$\Delta p = \frac{q}{2\pi\ r_f^2\ h\phi c\eta} \int_0^t \left[\int_0^{r_f} I_o \left(\frac{r_m\ r}{2\eta t'} \right) e^{-\frac{r^2}{4\eta t}}\ r\ d\ r \right]$$

$$e^{-\frac{r_m^2}{4\eta t'}}\ \frac{dt'}{t'} \qquad \text{(A-5)}$$

Using the dimensionless expressions defined by Eqs. 22, **23** and *24* in the text, Eq. **A-5** becomes:

$$p_D = \int^{t_D}\left[\int_0^1 I_o\left(\frac{r_D\, r_D'}{2t_D'}\right) e^{-\frac{r_D'^2}{4t_D}}\, r_D'\, dr_D'\right] e^{-\frac{r_D^2}{4t_D'}}\, dt_D' \tag{A-6}$$

Eq. A-6 represents the pressure drop function created by a solid cylinder source of radius r_f completely penetrating a reservoir of thickness h.

When $r_f \to 0$, it can be verified that the above solution reduces to the line source solution, when $r_D > 1$. We can write in Eq. A-5[36]:

$$I_o\left(\frac{r_m\, r}{2\eta t}\right) = 1 + \frac{r_m^2\, r^2}{16\,(\eta t)^2} + 0\,(r^4) \tag{A-7}$$

$$\exp\left(-\frac{r^2}{4\eta t}\right) = 1 - \frac{r^2}{4\eta t} + 0\,(r^4) \tag{A-8}$$

Thus:

$$\int_0^{r_f} I_o\left(\frac{r_m\, r}{2\eta t}\right)\exp\left(-\frac{r^2}{4\eta t}\right) r\, dr = \frac{r_f^2}{2} - \frac{r_f^4}{16\eta t} + \frac{r_f^4\, r_m^2}{4 \times 16\,(\eta t)^2} + 0\,(r_f^6) \tag{A-9}$$

Eq. A-5 becomes:

$$\Delta p = \frac{q_f}{2\pi h\phi c\eta} \frac{1}{2} \left\{ \int_0^t \frac{e^{-\frac{r_m^2}{4\eta t'}}}{t'} dt' - \frac{r_f^2}{16\eta} \int_0^t \frac{e^{-\frac{r_m^2}{4\eta t'}}}{t'^2} dt' \right.$$

$$\left. + \frac{r_f^2 r_m^2}{4 \times 16\eta^2} \int_0^t \frac{e^{-\frac{r_m^2}{4\eta t'}}}{t'^3} dt' + 0 (r_f^4) \right\} \qquad \text{(A-10)}$$

Using the new variable:

$$u = \frac{r_m^2}{4\eta t}$$

we can rewrite Eq. A-10 as:

$$\Delta p = \frac{q_f}{2\pi h\phi c\eta} \left\{ \frac{1}{2} \int_{\frac{r_m^2}{4\eta t}}^{\infty} \frac{e^{-u}}{u} du - \frac{r_f^2}{4r_m^2} e^{-\frac{r_m^2}{4\eta t}} + \frac{r_f^2}{16\eta t} e^{-\frac{r_m^2}{4\eta t}} \right.$$

$$\left. + \frac{r_f^2}{4r_m^2} e^{-\frac{r_m^2}{4\eta t}} + 0 (r_f^4) \right\} \qquad \text{(A-11)}$$

$$\Delta p = \frac{q_f}{2\pi h\phi c\eta} \left\{ -\frac{1}{2} Ei \left(-\frac{r_m^2}{4\eta t} \right) + \frac{r_f^2}{16\eta t} e^{-\frac{r_m^2}{4\eta t}} + 0 (r_f^4) \right\} \qquad \text{(A-12)}$$

Eq. A-5 (and therefore A-6) reduces to the line source solution when $r_m > r_f$, as $r_f \to 0$.

Long rime behavior:

The Laplace transform of p_D is obtained from the. Laplace transform of the P function:

$$L\{p_D\} = \frac{L\{P\}}{s} \tag{A-13}$$

An approximation for long time is obtained by taking the limit of Eq. A-13 as $s \to 0$.

a. $r_D > 1$

From Eq. 38:

$$L\{p_D\} = \frac{2K_o(r_D\sqrt{s})\, I_1(\sqrt{s})}{s\sqrt{s}} \tag{A-14}$$

As $s \to 0$[36],

$$I_1(\sqrt{s}) \sim \frac{\sqrt{s}}{2}$$

$$K_o(r_D\sqrt{s}) \sim -\left(\ell n\, \frac{r_D\sqrt{s}}{2} + \gamma\right)$$

and therefore:

$$L\{p_D\} \sim -\frac{1}{2}\,\frac{\ell n\ s}{s} - \frac{\gamma + \ell n\, \frac{r_D}{2}}{s} \tag{A-15}$$

The inverse transform of Eq. A-15 yields[34].

$$p_D = \frac{1}{2}(\ln t_D + \gamma) - \gamma - \ln \frac{r_D}{2} \tag{A-16}$$

and we obtain:

$$p_D = \frac{1}{2}\left(\ln \frac{t_D}{r_D^2} + 0.80907\right) \tag{A-17}$$

Eq. A-17 is the long time approximation of p_D $(r_D > 1, t_D)$ and is identical to the long time approximation of the line source. Therefore, the solid cylinder source is equivalent to a line source at large values of t_D for a given r_D. The same conclusion holds in fact for large r_D at any given value of t_D.

b. <u>$0 < r_D < 1$</u>

From Eq. 42:

$$L\{p_D\} = \frac{2}{s^2} - \frac{2I_o(r_D\sqrt{s})\, K_1(\sqrt{s})}{s\sqrt{s}} \tag{A-18}$$

as $z \to 0$ [36],
$$\begin{cases} K_1(z) = \frac{1}{z} + I_1(z)\left(\ln \frac{z}{2} + \gamma\right) - \frac{z}{4} + O(z^3) \\ I_1(z) = \frac{z}{2} + O(z^3) \\ I_o(z) = 1 + \frac{z^2}{4} + O(z^4) \end{cases}$$

Therefore, as $s \to 0$, Eq. A-18 can be approximated by:

$$L\{p_D\} = \frac{2}{s^2} - \frac{2}{s^2} - \frac{1}{s}\ln\frac{\sqrt{s}}{2} - \frac{\gamma}{s} + \frac{1}{2s} - \frac{r_D^2}{2s}$$

or:

$$L\,\{p_D\} = -\frac{1}{2}\,\frac{\ln s}{s} + \frac{1}{2s}\left(\ln 4 - 2\gamma + 1 - r_D^2\right)$$

The Laplace inverse of $-\frac{\ln s}{s}$ is $(\ln t_D + \gamma)$ and the inverse of p_D is:

$$p_D = \frac{1}{2}\left(\ln t_D + \ln 4 - \gamma + 1 - r_D^2\right)$$

or:

$$p_D = \frac{1}{2}\left(\ln t_D + 1.80907 - r_D^2\right) \qquad \text{(A-19)}$$

c. $\underline{r_D = 1}$

This case is a limiting case for both Eq. A-17 and A-19 and thus:

$$p_D = \frac{1}{2}\left(\ln t_D + 0.80907\right)$$

d. $\underline{r_D = 0}$

Integrating Eq. 44 from 0 to t_D and multiplying the result by 2, we obtain:

$$p_D = 2t_D\left(1 - e^{-\frac{1}{4t_D}}\right) - \frac{1}{2}\,\mathrm{Ei}\left(-\frac{1}{4t_D}\right) \qquad \text{(A-20)}$$

As $t_D \to \infty$:

$$p_D = 2t_D \left| 1 - 1 + \frac{1}{4t_D} + 0 \left(\frac{1}{t_D^2}\right) \right| + \frac{1}{2} (\ln t_D + 0.80907)$$

and the pressure drop function can be approximated by:

$$p_D = \frac{1}{2} (\ln t_D + 1.80907)$$

the result is identical to Eq. A-19, which thus applies for $0 \leqslant r_D \leqslant 1$.

Numerical values of the pressure drop function $p_D(r_D, t_D)$ are given in Table I for different values of r_D, as time increases.

TABLE 1 - Table of $p_D(r_D,\ t_D)$ Values

t_D \ r_D	0.00	0.25	0.50	0.75	1.00	2.00	3.00	5.00	10.00
$1.^{-2}$	0.0200	0.0200	0.0200	0.0197	0.0096				
2.	0.0400	0.0400	0.0399	0.0380	0.0189				
3.	0.0600	0.0600	0.0595	0.0547	0.0280				
4.	0.0800	0.0799	0.0787	0.0705	0.0370				
6.	0.1197	0.1191	0.1151	0.0996	0.0544				
8.10	0.1585	0.1570	0.1494	0.1265	0.0714	0.0001			
$1.^{-1}$	0.1960	0.1932	0.1817	0.1517	0.0879	0.0004			
2.10^{-1}	0.3586	0.3494	0.3194	0.2614	0.1651	0.0047			
3.	0.4855	0.4718	0.4289	0.3521	0.2343	0.0141	0.0003		
4.10^{-1}	0.5879	0.5711	0.5194	0.4296	0.2965	0.0272	0.0015		
6.	0.7466	0.7259	0.6633	0.5568	0.4036	0.0597	0.0055		
8.10	0.8673	0.8444	0.7753	0.6587	0.4930	0.0958	0.0134		
1	0.9646	0.9402	0.8668	0.7437	0.5693	0.1327	0.0243	0.0003	
2	1.2820	1.2544	1.1715	1.0332	0.8389	0.2986	0.0994	0.0070	
3	1.4747	1.4459	1.3596	1.2156	1.0136	0.4292	0.1799	0.0244	
4	1.6135	1.5841	1.4960	1.3491	1.1432	0.5342	0.2541	0.0489	0.0002
6	1.8111	1.7811	1.6912	1.5412	1.3312	0.6960	0.3802	0.1060	0.0017
8	1.9524	1.9221	1.8312	1.6797	1.4675	0.8186	0.4828	0.1642	0.0059
10	2.0624	2.0319	1.9405	1.7880	1.5745	0.9171	0.5686	0.2195	0.0130
20	2.4062	2.3753	2.2827	2.1283	1.9122	1.2372	0.8614	0.4401	0.0741
30	2.6080	2.5769	2.4839	2.3289	2.1119	1.4308	1.0454	0.5964	0.1471
40	2.7513	2.7202	2.6270	2.4717	2.2542	1.5701	1.1798	0.7161	0.2169
60	2,9535	2.9224	2.8290	2.6733	2.4554	1.7682	1.3729	0.8940	0.3383
80	3.0971	3.0659	2.9724	2.8166	2.5985	1.9097	1.5119	1.0252	0.4383
1.10^{2}	3.2085	3.1773	3.0838	2.9277	2.7096	2.0199	1.6205	1.1291	0.5226
2.10^{2}	3.5551	3.5238	3.3402	3.2741	3.0555	2.3640	1.9615	1.4604	0.8121
3.10^{2}	3.7578	3.7265	3.6328	3.4767	3.2580	2.5658	2.1623	1.6580	0.9950
4.10^{2}	3.9016	3.8703	3.7766	3.6204	3.4017	2.7092	2.3052	1.7992	1.1288
6.10^{2}	4.1043	4.0730	3.9793	3.8230	3.6043	2.9115	2.5070	1.9993	1.3213
8.10^{2}	4.2481	4.2168	4.1231	3.9668	3.7481	3.0551	2.6503	2.1418	1.4600
1.10^{3}	4.3596	4.3284	4.2346	4.0784	3.8596	3.1666	2.7616	2.2526	1.5685

APPENDIX B

Unsteady-State Pressure Distribution Created by a Single Plane Horizontal Fracture (fracture of zero--thickness)

We will derive the solution directly using the superposition principle as used previously for the solid cylinder,, and the fracture af finite thickness. We start with Eq. 7 in the text, in which we set $z = z_f$, and we integrate with respect to r from 0 to r_f to generate an instantaneous disc. The result is:

$$\Delta p = \frac{q\, dt}{4\phi c\sqrt{\pi}\,(\eta t)^{3/2}} \exp\left[-\frac{r_m^2 + (z_m - z_f)^2}{4\eta t}\right] \int_0^{r_f} I_o\left(\frac{r_m r}{2\eta t}\right) e^{-\frac{r^2}{4\eta t}}\, r\, dr \qquad \text{(B-1)}$$

The constant withdrawal rate is uniform over the surface of the fracture and is given by:

$$q_f = \pi r_f^2 q \qquad \text{(B-2)}$$

We follow the same procedure as used in Chapter I to generate the upper and lower reservoir boundaries. The pressure drop created by the disc in a semi infinite medium bounded by the lower boundary is given by:

$$\Delta p = \frac{q_f\, dt}{4r_f^2\, \phi c\, (\pi\eta t)^{3/2}}\, e^{-\frac{r_m^2}{4\eta t}} \left\{ \exp\left[-\frac{(z_m - z_f)^2}{4\eta t}\right] + \exp\left[-\frac{(z_m + z_f)^2}{4\eta t}\right] \right\} \int_0^{r_f} I_o\left(\frac{r_m\, r}{2\eta t}\right) e^{-\frac{r^2}{4\eta t}}\, r\, dr \tag{B-3}$$

The instantaneous pressure drop created by the plane fracture in the actual reservoir is then given by:

$$\Delta p = \frac{q_f\, dt}{4r_f^2\, \phi c\, (\pi\eta t)^{3/2}}\, e^{-\frac{r_m^2}{4\eta t}} \sum_{n=-\infty}^{+\infty} \left\{ \exp\left[-\frac{(z_m - z_f - 2nh)^2}{4\eta t}\right] + \exp\left[-\frac{(z_m + z_f - 2nh)^2}{4\eta t}\right] \right\} \int_0^{r_f} I_o\left(\frac{r_m\, r}{2\eta t}\right) e^{-\frac{r^2}{4\eta t}}\, r\, dr \tag{B-4}$$

As mentioned in Chapter I, we can write:

$$\sum_{n=-\infty}^{+\infty} \exp\left[-\frac{(z_m - z_f - 2nh)^2}{4\eta t}\right] =$$

$$\frac{\sqrt{\pi\eta t}}{h} \left[1 + 2 \sum_{n=1}^{\infty} \cos n\pi\, \frac{z_m - z_f}{h} \exp\left(-\frac{n^2\pi^2\eta t}{h^2}\right)\right] \tag{B-5}$$

$$\sum_{n=-\infty}^{+\infty} \exp\left[-\frac{(z_m + z_f - 2nh)^2}{4\eta t}\right] =$$

$$\frac{\sqrt{\pi\eta t}}{h}\left[1 + 2\sum_{n=1}^{\infty} \cos n\pi \frac{z_m + z_f}{h} \exp\left(-\frac{n^2\pi^2\eta t}{h^2}\right)\right] \qquad \text{(B-6)}$$

therefore:

$$\sum_{n=-\infty}^{+\infty}\left\{\exp\left[-\frac{(z_m - z_f - 2nh)^2}{4\eta t}\right] + \exp\left[-\frac{(z_m + z_f - 2nh)^2}{4\eta t}\right]\right\}$$

$$= \frac{2\sqrt{\pi\eta t}}{h}\left[1 + 2\sum_{n=1}^{\infty} \exp\left(-\frac{n^2\pi^2\eta t}{h^2}\right) \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z_m}{h}\right] \qquad \text{(B-7)}$$

and Eq. B-4 becomes:

$$\Delta p = \frac{q_f \, dt}{2\pi r_f^2 \, h\phi c\eta t} \, e^{-\frac{r_m^2}{4\eta t}} \left[1 + 2\sum_{n=1}^{\infty} \exp\left(-\frac{n^2\pi^2\eta t}{h^2}\right) \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z_m}{h}\right]$$

$$\int_0^{r_f} I_o\left(\frac{r_m r}{2\eta t}\right) e^{-\frac{r^2}{4\eta t}} \, r \, dr \qquad \text{(B-8)}$$

The pressure drop created by a continuous plane fracture is then given by:

$$\Delta p = \frac{q_f}{2\pi r_f^2 \, h\phi c\eta}$$

$$\int_0^t \left[1 + 2 \sum_{n=1}^{\infty} \exp\left(- \frac{n^2\pi^2\eta(t - t')}{h^2} \right) \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z_m}{h} \right]$$

$$\left[\int^{r_f} I_o \left(\frac{r_m \, r}{2\eta \, (t - t')} \right) e^{- \frac{r^2}{4\eta \, (t - t')}} \, r \, dr \right] \frac{e^{- \frac{r_m^2}{4\eta(t - t')}}}{t - t'} \, dt'$$

(B-9)

which again can be simplified to:

$$\Delta p = \frac{q_f}{2\pi r_f^2 \, h\phi c\eta} \int_0^t \left[1 + 2 \sum_{n=1}^{\infty} \exp\left(- \frac{n^2\pi^2\eta t'}{h^2} \right) \cos n\pi \, {}^{z} f \cos n\pi \frac{z_m}{h} \right]$$

$$\left[\int_0^{r_f} I_o \left(\frac{r_m \, r}{2\eta t'} \right) e^{- \frac{r^2}{4\eta t'}} \, r \, dr \right] \frac{e^{- \frac{r_m^2}{4\eta t'}}}{t'} \, dt \qquad \text{(B-10)}$$

Using the dimensionless expressions defined by Eqs. 22 to 25, we obtain:

$$p_D(r_D, z_D, t_D) = \int_0^{t_D} \left[1 + 2 \sum_{n=1}^{\infty} e^{-\frac{n^2 \pi^2 t_D'}{h_D^2}} \cos n\pi \frac{z_f}{h} \cos n\pi \frac{z}{h} \right]$$

$$\left[\int_0^1 I_o \left(\frac{r_D \; r_D'}{2t_D'} \right) e^{-\frac{r_D'^2}{4t_D'}} r_D' \, dr_D' \right] \frac{e^{-\frac{r_D^2}{4t_D'}}}{t_D'} \, dt_D' \qquad \text{(B-11)}$$

Eq. B-11 is identical to Eq. 133 in the text, which has been derived from the solution for a fracture of finite thickness.

APPENDIX C

Unsteady-State Pressure Distribution in Linear Systems with Constant Influx and Wellbore Storage Effects

We consider vertical linear flow at a constant rate $\frac{q_f}{2}$ into a horizontal fracture which has a storage constant equal to C. At early times, we can assume the horizontal reservoir boundaries do not affect the solutfon (infinite case).

The mathematical expression of the problem is as follows:

$$\frac{d^2p}{dz'^2} = \frac{\phi\mu c}{k_z} \frac{dp}{dt} \tag{C-1}$$

with the conditions:

$$p = p_i \begin{cases} t = 0 \text{ all } z' \\ t > 0 \; z' = \infty \end{cases} \tag{C-2}$$

$$\frac{dP}{dz'} = \frac{q_f \mu}{2\pi r_f^2 k} + C \frac{dP}{dt} \quad t > 0 \text{ and } z' = 0 \tag{C-3}$$

where $z' = z - (z_f + \frac{h_f}{2})$ above the fracture, or $z' = (z_f - \frac{h_f}{2}) - z$ below. We define the following dimensionless quantities:

$$p_D' = \frac{2\pi r_f \sqrt{k_r k_z}}{q_f \mu} (p_i - p) = \frac{p_D}{h_D} \tag{C-4}$$

$$z'_D = \frac{z'}{r_f}\sqrt{\frac{k_r}{k_z}} \tag{C-5}$$

$$t_D = \frac{k_r\ t}{\phi\mu c\ r_f^2} \tag{C-7}$$

$$\bar{C} = \frac{\sqrt{k_r\ k_z}}{\phi\mu c\ r_f}\ C \tag{C-8}$$

The equations defining the problem become:

$$\frac{d^2 p'_D}{dz'^2_D} = \frac{dp_D}{dt_D} \tag{C-9}$$

$$p'_D = 0 \begin{cases} t_D = 0 & \text{all } z_] \\ t_D > 0 & z'_D = \text{‹} \end{cases} \tag{C-10}$$

$$\frac{dp'_D}{dz_D} = -1 + \bar{C}\,\frac{dp'_D}{dt_D} \quad t_D > 0 \text{ and } z'_D = 0 \tag{C-11}$$

Applying the Laplace transform to Eqs. C-9 to C-11, we obtain:

$$\frac{d^2 L\ \{p'_D\}}{dz'^2_D} - s\ L\ \{p'_D\} = 0 \tag{c-12}$$

$$\left.\frac{d\,L\,\{p_D'\}}{dz_D'}\right|_{z_D'=0} = -\frac{1}{s} + \bar{C}\,s\,L\,\{p_D'\}\Big|_{z_D'=0} \tag{C-13}$$

$$L\,\{p_D'\} = 0 \quad z_D' = \infty \tag{C-14}$$

The solution of the system defined by Eq. (C-12) to (C-14) is then:

$$L\,\{p_D'\} = \frac{e^{-z_D'\sqrt{s}}}{s^{3/2}(1 + \bar{C}\sqrt{s})} \tag{C-15}$$

which, at early times is equivalent to:

$$L\,\{p_D'\} = \frac{e^{-z_D'\sqrt{s}}}{\bar{C}\,s^2} \tag{C-16}$$

which is similar to Eqs. 121 and 125 in the text.

At $z_D' = 0$, the solution reduces to:

$$p_D' = \frac{t_D}{\bar{C}} \tag{C-17}$$

or:

$$p_D = \frac{h_D}{\bar{C}}\;t_D \tag{C-18}$$

A similar treatment of this problem has been given by Carslaw and Jaeger[49] for the finite case.

APPENDIX D

Single Plane Horizontal Fracture Case - Computational Procedure and Numerical Results

The solid cylinder source solution has been first evaluated from Eq. 31 which has been written in the form:

$$p_D(r_D, t_D) = \int_0^{t_D} \left[\int_0^1 \exp\left(-\frac{(r_D - r_D')^2}{4t_D'}\right) \exp\left(-\frac{r_D\, r_D'}{2t_D'}\right) I_o\left(\frac{r_D\, r_D'}{2t_D'}\right) r_D'\, dr_D' \right] \frac{dt_D'}{t_D'} \qquad \text{(D-1)}$$

in order to avoid positive arguments in the exponential functions because the product $\left[\exp\left(-\frac{r_D\, r_D'}{2t_D'}\right) I_o\left(\frac{r_D\, r_D'}{2t_D'}\right)\right]$ tends to $\sqrt{\frac{t_D'}{r_D\, r_D'}}$ when t_D becomes small[36].

The term between brackets has been evaluated for each value of time using Romberg's rule[50] over the interval $0 \leqslant r_D' \leqslant 1$, which has been divided into as many as *640* equal subintervals. This number was imposed by storage considerations with the IBM 360/67 used for the computations.

Eq. D-1 was then evaluated using the trapezoidal rule. The accuracy of the results was checked in the following manner:

(1) for $r_D = 0$, the results were compared with the analytical solution given by Eq. A-20 in Appendix A. Several attempts were necessary to provide good accuracy. The calculations were finally made with 90 values of time, t_D, per log cycle. Because of the

properties of Romberg's rule, the calculated solution converges towards the exact solution as the number of points used in the computation increases.

(2) for $r_D > 1$ and large values of time, a comparison can be made with solutions obtained by using the Laplace transform[51] tabulated by M. J. Edwardson, et al.[52]. The difference is less than 0.1%. The results for the solid cylinder are shown in Table 1 for different values of r_D.

The pressure created by the plane fracture was obtained by adding the pseudo skin function to the solid cylinder source solution as in Eq. 137. The particular properties of the P and Z functions discussed in the main body of this study, were used in the computations. The results of the computations are shown in Table 2, where the dimensionless wellbore pressure is given as a function of dimensionless time for an extensive range of values of the dimensionless formation thickness h_D. Some results have been plotted in Fig. 11 on log-log coordinates to show the early time behavior, and in Fig. 12 on semi-log coordinates to show the transient flow period.

Variation of the pressure in the fracture for a given value of dimensionless formation thickness is illustrated in Figs. 13 and 14.

Table 2. Table of $p_D(r_D, z_D, t_D)$ values for a single plane horizontal fracture at center of formation. for various dimensionless reservoir thicknesses, h_D ($r_D = 0.0$; $z_D = 0.5$)

t_D \ h_D	0.05	0.2	0.3	0.4	0.5	0.7	1.0	2.0	3.0	4.0	5.0	6.0	7.0	10.0
1.10^{-2}	[$p_D(0,t_D)$* +0.0004]	[$p_D(0,t_D)$ +0.0[illegible]]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2.10^{-2}			[$p_D(0,t_D)$ +0.0150]	0.0666	[illegible]	[illegible]	[illegible]	[illegible]	0.4788	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
3.10^{-2}				[$p_D(0,t_D)$ +0.0267]	0.1015	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
4.10^{-2}					[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	2.2570
6.10^{-2}					[$p_D(0,t_D)$ +0.0417]	[illegible]	[illegible]	[illegible]	0.8281	[illegible]	[illegible]	[illegible]	[illegible]	2.7605
8.10^{-2}						[illegible]	0.3207	0.6352	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
1.10^{-1}						[$p_D(0,t_D)$ +0.0817]	0.3604	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	3.5293
2.10^{-2}							[$p_D(0,t_D)$ +0.1661]	[illegible]	[illegible]	1.8960	[illegible]	[illegible]	[illegible]	4.7401
3.10^{-1}								[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
4.10^{-1}								[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
6.10^{-1}								[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
8.10^{-1}								1.4903	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
1								[$p_D(0,t_D)$ +0.6231]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2									[$p_D(0,t_D)$ +1.[illegible]40]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
3										[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
4										[$p_D(0,t_D)$ +1.9802]	4.3771	5.1955	6.0375	[illegible]
6											[$p_D(0,t_D)$ +2.763[illegible]]	5.3963	6.2434	[illegible]
8												[illegible]	[illegible]	[illegible]
10												[$p_D(0,t_D)$ +3.5854]	6.4960	9.1395
20													[$p_D(0,t_D)$ +4.4337]	[$p_D(0,t_D)$ +7.[illegible]]
30														

*$p_D(0,t_D)$ is given in Table 1.

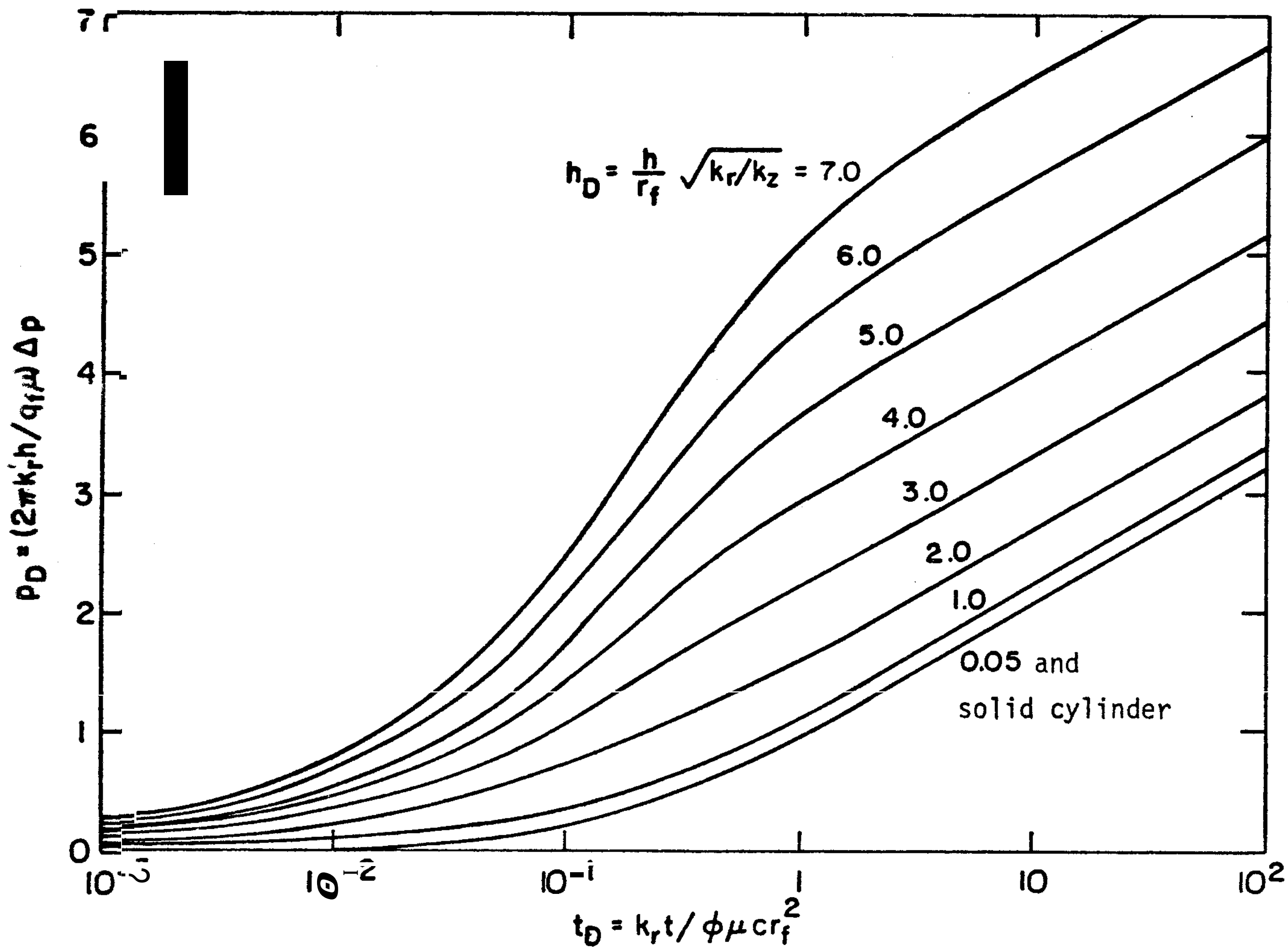

Fig. 12. Dimensionless pressure in the fracture vs. dimensionless time for a single plane horizontal fracture at center of formation (r_D = 0.0)

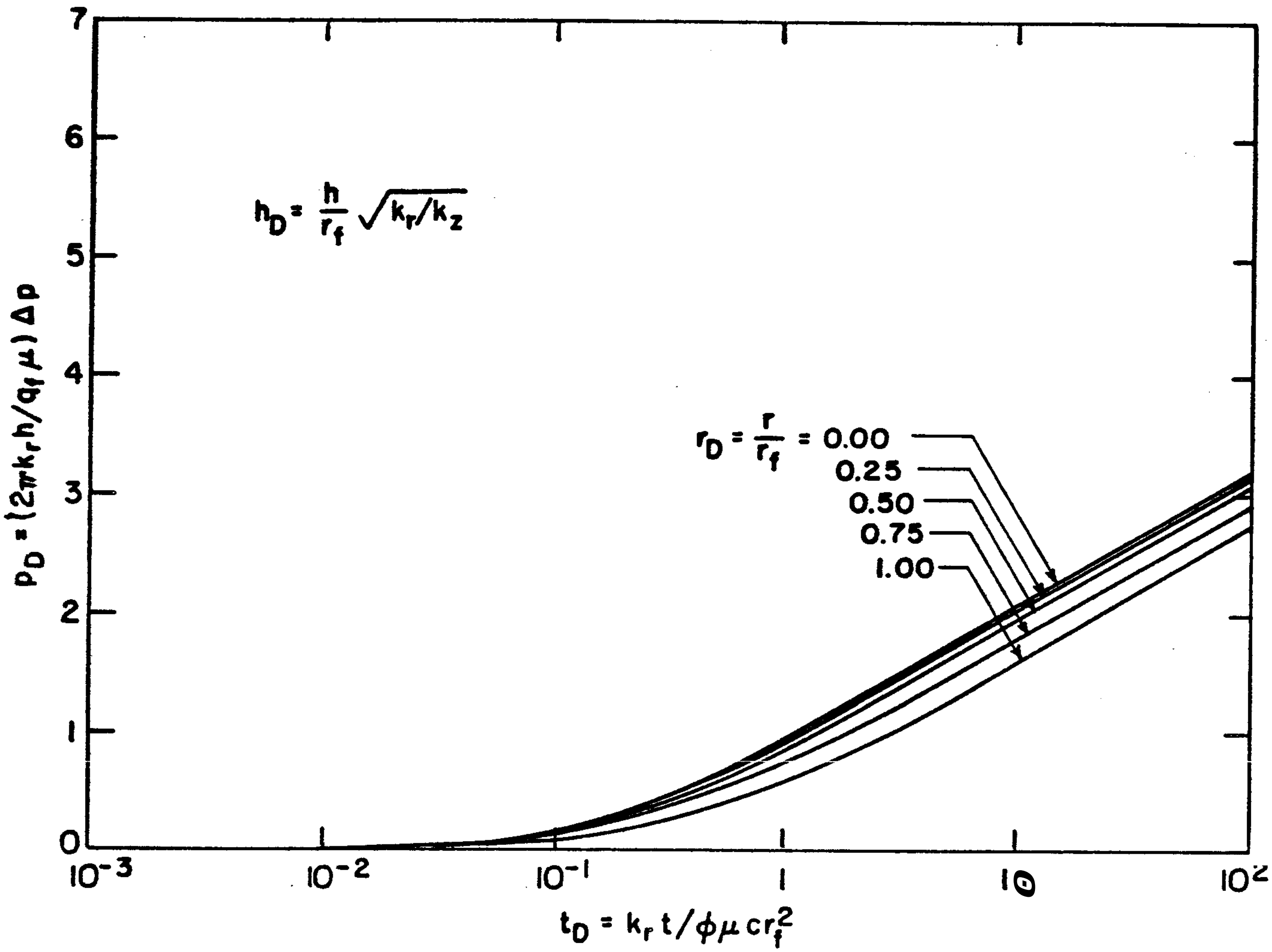

Fig. 13. Dimensionless pressure in the fracture vs. dimensionless time for a single plane horizontal fracture at center of formation (h_D = 0.05)

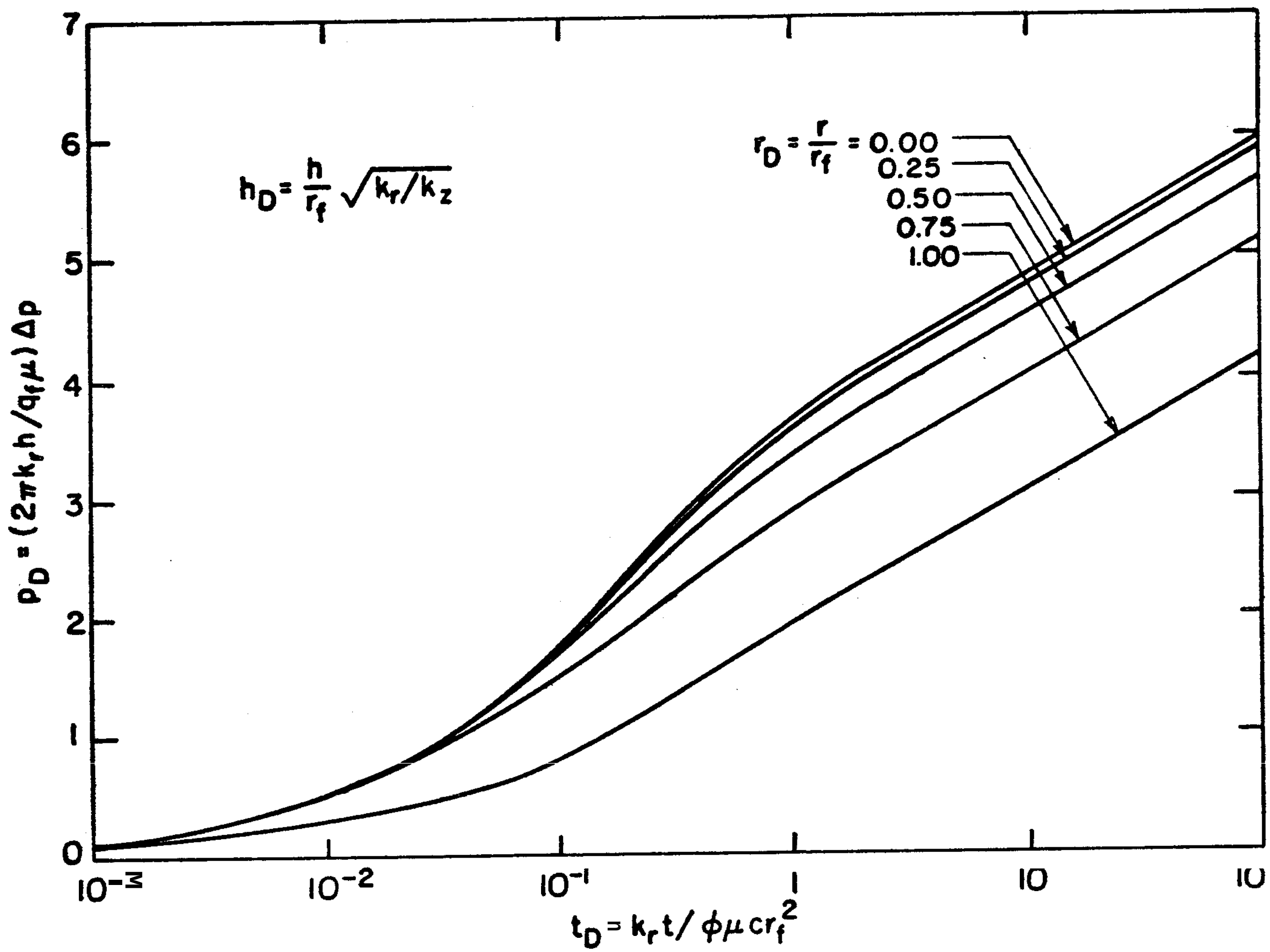

Fig 14 Dimensionless pressure in the fracture vs. dimensionless time for a single plane horizontal fracture at center of formation (h_D = 5.0)